

Exact Stieltjes and Harmonic Identities

Integer dilations, three-factor products,
Gamma coefficients, and corrected conjectures

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Abstract

We develop exact identities extending the periodic Hurwitz–Stieltjes calculus in the ProveIt manuscript and its incoming research reports. For two affine integer frequencies we prove a complete reduction to the unit-frequency shifted correlation, including every Stieltjes index and the finite polynomial that records changes of local cutoff scale and argument differentiation. For three distinct translations we give a six-term colored Tornheim formula, prove its entire kernel continuation, and extract all finite-part Stieltjes products and ordinary correlations of their normalized antiderivatives. A separate Gamma-quotient calculus reduces shifted elementary harmonic sums with arbitrary convergent rational kernels to finite polygamma expressions and determines their complete principal parts and constants at the moving spectral pole. Gauss summation supplies convergent series for every generalized Stieltjes constant, all mixed parameter derivatives, and successive negative-integer resonances. Finally, one generating identity proves corrected versions of two displayed harmonic conjecture families in a 2014 paper and identifies the accompanying normalization and harmonic-order errors. Every principal result has an analytic proof. Exact symbolic generators, independent numerical diagnostics, a source audit, and a prioritized research agenda accompany the article. Classical ingredients and results already in the repository are explicitly distinguished from the present extensions; historical priority is not asserted.

Repository snapshot: `fc4d3bf80534ad7c901d3b8c9e71baf2df0064ed`.

Research manuscript for review and integration; no upstream files have been changed.

Contents

1	Scope, results, and relation to the source material	1
1.1	Main outcomes	1
1.2	What is classical, what is extended, and what remains open	1
1.3	Reading and integration order	2
2	Analytic conventions and finite parts	2
2.1	Periodic distributions and the endpoint coordinate	2
2.2	Products, regular coefficients, and legitimate exchanges	3
3	Integer dilations and exact local scale corrections	3
3.1	The reduced frequency and shifted spectral kernel	4
3.2	A finite polynomial for every scale and derivative	4
3.3	A self-contained Gamma quotient for the base correlation	6
3.4	All Stieltjes indices in an explicit finite formula	7
3.5	An ordinary log-Gamma identity with unequal frequencies	8
3.6	Audit finding and scope	9
4	Three translated factors and colored Tornheim jets	9
4.1	A meromorphic identity before coefficient extraction	10
4.2	The precise endpoint finite part	11
4.3	An explicit formula for three digamma factors	12
4.4	A convergent representation of the first spectral derivative	13
4.5	Integral inner orders and multiple polylogarithms	13
4.6	Ordinary triple log-Gamma correlations	13
5	A Gamma coefficient calculus for shifted harmonic sums	14
5.1	An exact generating theorem	15
5.2	Explicit polygamma reductions and antiderivatives	16
5.3	Closure under arbitrary rational kernels	17
5.4	The moving pole and all-depth finite parts	18
5.5	Stieltjes constants, the literature comparison, and limits of the reduction	19
6	Gauss summation at zero balance and Stieltjes jets	20
6.1	A convergent master identity	21
6.2	Every spectral derivative in Bell-polynomial form	22
6.3	A complete mixed-parameter coefficient rule	22
6.4	Ordinary and central-binomial Stieltjes series	23
6.5	Polylogarithmic boundary values	24
7	Successive subtractions and every negative balance	25
7.1	An exact jet formula at a negative integer	26
7.2	An exact central-binomial identity at balance minus one	27
8	Correcting and proving a historical family of harmonic conjectures	28

9	Source audit, proof checks, and reproducibility	29
9.1	The five incoming reports	29
9.2	Proposed source corrections	30
9.3	The surviving named Gaussian conjectures	30
9.4	Independent numerical diagnostics	31
9.5	Exact arithmetic and reproducible artifacts	32
10	Further research questions and concrete next targets	32
10.1	The frozen Gaussian relations at weights seven and nine	33
10.2	Minimal coordinates for the first triple Stieltjes periods	33
10.3	Three unequal integer frequencies	33
10.4	Collision identities and nonlinear coordinate changes	33
10.5	Mixed harmonic Stieltjes data beyond the constant term	34
10.6	Negative-balance identities and other summation theorems	34
10.7	The full polylogarithmic boundary function	34
10.8	Exact symbolic generation and formal proof interfaces	34
A	Exact coefficient checks and explicit correction polynomials	35
A.1	The leading triple pole	35
A.2	First bilinear closure coefficients	35
A.3	The fifth and sixth correction polynomials	35

1 Scope, results, and relation to the source material

This article continues two explicitly identified sources: the ProveIt polylogarithm manuscript and its incoming research directory [1, 2]. The repository was examined at commit `fc4d3bf80534ad7c901d3b8c9e71baf2df0064ed`. The five incoming ZIP reports dated 10 October 2026 are distinguished individually in the source manifest and in Section 9. Representative manuscript chapters and the relevant incoming sections were read for definitions, proof dependencies, and open questions. This is a targeted mathematical audit, not a certification of every statement in the 417-page source manuscript.

The source material already contains extensive one-factor differentiation and antiderivative formulas, unit-frequency bilinear Stieltjes closure, and log-Gamma correlations. Repeating those as new discoveries would misstate the problem. The most useful next steps are the exact behavior under unequal integer dilations, the first genuinely double-index correlation problem with three translated factors, and independent coefficient constructions joining harmonic series to Stieltjes constants.

1.1 Main outcomes

The following list describes what is proved, with the hypotheses that matter for integration into the repository.

1. **Two affine frequencies.** For $p = dP$, $q = dQ$, $\gcd(P, Q) = 1$, and $c = \{Pb - Qa\} \neq 0$, the entire ordinary two-factor Hurwitz family reduces to $Q^{s-1}P^{t-1}C(s, t; c)$. All Stieltjes coefficients then reduce to finitely many $\gamma_j(c)$ and $\gamma_j(1-c)$. The universal polynomial $n![z^{n+1}]e^{Lz} \prod_{j=1}^k (1 - z/j)$ supplies every local scale and differentiation correction. See Theorems 3.1, 3.2, and 3.4.
2. **Three translated singular factors.** When the three shifts are pairwise distinct, the spectral integral is the sum of six explicitly colored Tornheim kernels. A holomorphic coefficient formula gives every finite-part triple Stieltjes product, and specialization gives three digammas and correlations of arbitrary normalized Stieltjes primitives. See Theorems 4.2 and 4.4, and Corollary 4.5. The coordinate class is proved; a reduction of every Tornheim jet to single-variable Stieltjes constants is not claimed.
3. **Harmonic sums and their endpoint constants.** If $E_{n,r}$ is the elementary harmonic coefficient of depth r , then every convergent sum $\sum_n E_{n,r} R(n)$ with rational R and permitted poles has an explicit finite Gamma-coefficient reduction. A jointly holomorphic remainder at the moving pole determines the principal part and constant at every depth. See Theorem 5.1, Corollary 5.2, and Theorem 5.3.
4. **Gauss summation and all Stieltjes orders.** Subtracting the matching Hurwitz term from a gamma ratio gives a normally convergent series through zero balance. Its arbitrary mixed parameter jets are finite Bell-polynomial expressions with the correct generalized Stieltjes term. Further asymptotic subtractions give exact continuation and residue identities at each negative integer. See Theorems 6.1, 6.2, and 6.3.
5. **Corrected harmonic conjectures.** A single two-parameter Gamma quotient proves all orders of the harmonic identities underlying Conjectures 4 and 6 in Choi's inspected 2014 paper [16]. It also corrects Corollary 5: the displayed right sides (41)–(46) have an extra factor two, and two harmonic-order misprints require repair. See Theorem 8.1. These are statements about the printed formulas, without any assertion that the corrected identities remained open in 2026.

1.2 What is classical, what is extended, and what remains open

The organizing tools are classical: the Hurwitz Fourier formula, Gamma reflection and Gauss summation, elementary harmonic generating functions, and Mellin continuation by Taylor subtraction

[3, 14]. Mikolás’s dilation formula, as recorded by Espinosa and Moll, is an antecedent to the frequency calculation [5]. Their Tornheim work and the colored reductions of Zhao and Zhao–Zhou are relevant antecedents to the triple-product coordinate class [6, 7, 8]. The unshifted height-one Gamma generating function and harmonic Stieltjes constants likewise have a substantial existing literature [11, 9, 12].

The additions established here combine those tools with explicit shifts, all parameter orders, and stated finite-part coordinates, and prove the resulting formulas directly. Such an exact extension can answer a repository research question even when its underlying method is classical. No global novelty claim follows merely from not finding an identical formula in the reviewed sources.

The surviving manuscript problems called S6 and S8 are not declared proved by this article. Their narrower coefficient-algebra and function-class requirements are separate from an exact representation in a larger special-function class. The same distinction applies to coincident singularities, unrestricted higher harmonic Stieltjes data, and arithmetic independence. Section 10 records these limits and concrete next questions.

1.3 Reading and integration order

Section 2 fixes signs, Fourier phases, and the meaning of a finite part. Sections 3 and 4 form the periodic-correlation continuation. The harmonic and Gauss sections can be read independently after the Stieltjes convention is fixed. The proofs use generating identities before coefficient extraction; this makes arbitrary orders reproducible without listing hundreds of special cases. The accompanying Python programs implement exact coefficient arithmetic and independent numerical comparisons. None of the analytic conclusions rests on floating-point agreement.

2 Analytic conventions and finite parts

For $\Re a > 0$, the Hurwitz zeta function has the normalization

$$\zeta(1+u, a) = \frac{1}{u} + \sum_{m=0}^{\infty} \frac{(-1)^m \gamma_m(a)}{m!} u^m, \quad \gamma_0(a) = -\psi(a). \quad (2.1)$$

Here $\psi = \Gamma'/\Gamma$, $\psi^{(j)}$ denotes an argument derivative, and $\gamma = \gamma_0(1)$ is Euler’s constant. A prime on $\zeta(s, a)$ denotes an order derivative unless its variable is explicitly stated. The same convention applies to $\partial_s \text{Li}_s(z)$. Powers of positive real numbers use real logarithms; powers of $n+a$ in the right half-plane use the principal logarithm. At a nontrivial unit-circle twist, the polylogarithm is the continuation from the open unit disk [3, 4].

The basic argument identities used repeatedly below are

$$\partial_a^k \zeta(s, a) = (-1)^k (s)_k \zeta(s+k, a), \quad (2.2)$$

$$\psi^{(j)}(a) = (-1)^{j+1} j! \zeta(j+1, a), \quad j \geq 1, \quad (2.3)$$

$$\zeta'(0, a) = \log \Gamma(a) - \frac{1}{2} \log(2\pi). \quad (2.4)$$

These identities, with meromorphic continuation where needed, are standard [3]. In particular, $\gamma_m(t) = t^{-1} \log^m t + \gamma_m(1+t)$, which identifies the complete leading endpoint singularity of each Stieltjes function.

2.1 Periodic distributions and the endpoint coordinate

Identify the unit circle with $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, and write $\{x\}$ for the fractional part away from its isolated endpoints. For $\Re s < 1$ define $Z(s)$ by the locally integrable periodic function $\zeta(s, \{x\})$. Its Fourier coefficient convention is $\hat{T}(n) = \langle T, e^{-2\pi i n x} \rangle$. For $n \neq 0$,

$$\hat{Z}(1-z)(n) = \Gamma(z)(2\pi|n|)^{-z} e^{-i\pi z \operatorname{sgn}(n)/2}, \quad \hat{Z}(1-z)(0) = 0. \quad (2.5)$$

The formula first holds in an ordinary convergent domain and then as a meromorphic distribution identity [3].

For clarity, its local expansion at $z = 0$ is

$$Z(1 - z) = \frac{\delta_0 - 1}{z} + \sum_{m \geq 0} \frac{G_m}{m!} z^m. \quad (2.6)$$

Here 1 is the constant distribution, δ_0 has unit mass, and

$$\langle G_m, \phi \rangle = \text{CT}_{\varepsilon \downarrow 0} \int_{\varepsilon}^1 \gamma_m(x) \phi(x) dx \quad (2.7)$$

for a smooth periodic test function ϕ . The symbol CT retains the constant term after removing divergent powers and nonconstant logarithmic terms. The cutoff here is the ordinary coordinate x . To verify (2.6), split $\zeta(1 - z, x) = x^{z-1} + \zeta(1 - z, 1 + x)$ and write

$$\int_0^1 x^{z-1} \phi(x) dx = \frac{\phi(0)}{z} + \int_0^1 x^{z-1} (\phi(x) - \phi(0)) dx.$$

The second integral is holomorphic near zero. The other Hurwitz term contributes $-z^{-1} \int \phi$ and its ordinary regular coefficients. Expansion gives exactly (2.7). This also proves $\langle G_m, 1 \rangle = 0$, either directly or from the zero Fourier coefficient.

At a translated singularity x_0 , a fixed-coordinate finite part means the right coordinate $t = x - x_0$ on the circle. The coordinates t and pt give different finite constants. For example,

$$\text{CT}_{\varepsilon \downarrow 0} \int_{\varepsilon}^h \frac{\log^m(pt)}{pt} dt = \frac{\log^{m+1}(ph) - \log^{m+1}p}{p(m+1)}.$$

The explicit second term is the source of the scale corrections in Section 3. A spectral finite part is a different operation: it is a Laurent coefficient in a spectral variable. The article relates the two by local continuation and does not silently identify them.

2.2 Products, regular coefficients, and legitimate exchanges

If periodic factors have disjoint singular supports, their product as distributions is defined locally by multiplying a single singular distribution by smooth functions. A partition of unity makes this definition global and independent of that partition. This elementary criterion applies to the separated two- and three-factor products in this article. It does not define a product at a collision.

Spectral coefficient extraction is justified after endpoint subtraction, on compact parameter sets. A local term $t^{z-1}g(t)$ is replaced by $g(0)h^z/z + \int_0^h t^{z-1}(g(t) - g(0)) dt$; the remainder has an integrable uniform majorant. At simultaneous parameter poles, we multiply by the corresponding pole factors and use Cauchy's formula on a closed polydisk. For positive derivative orders in the local coordinate, a higher Taylor polynomial is subtracted. These are analytic arguments for exchanging limits, derivatives, and integrals; no divergent series is rearranged by formal cancellation alone.

For a formal holomorphic series, $[u^k v^\ell]F$ always denotes the ordinary Taylor coefficient, without factorials. Thus $\partial_u^k \partial_v^\ell F(0, 0) = k! \ell! [u^k v^\ell]F$. The factorial convention is stated again whenever two common harmonic normalizations occur together.

3 Integer dilations and exact local scale corrections

Two unequal integer frequencies still leave a single free Fourier index. This elementary fact resolves the unequal-dilation question in the incoming correlation reports. The spectral reduction is classical in method: Espinosa–Moll, Part 1, equation (1.13), already records an unshifted, equal-order dilation formula due to Mikolás. The additional issue here is the finite part in an explicitly fixed local coordinate, together with all spectral and argument derivatives. No historical priority claim is made for the underlying Fourier factorization.

3.1 The reduced frequency and shifted spectral kernel

Let $p, q \geq 1$ be integers, let

$$d = \gcd(p, q), \quad p = dP, \quad q = dQ, \quad \gcd(P, Q) = 1, \quad c = \{Pb - Qa\}.$$

The shifts a, b are real and henceforth $0 < c < 1$. This condition is exactly that the singular sets of the two periodic factors are disjoint. Indeed, simultaneous singularities would give $px + a = j$, $qx + b = k$ and hence $Pb - Qa = Pk - Qj \in \mathbb{Z}$; conversely, Bezout's identity gives such a solution whenever this expression is integral.

Write

$$C(s, t; c) = \int_0^1 \zeta(s, x) \zeta(t, \{x + c\}) dx, \quad \Re s, \Re t < 1.$$

Theorem 3.1 (Complete bilinear integer-dilation reduction). *For $\Re s, \Re t < 1$,*

$$\int_0^1 \zeta(s, \{px + a\}) \zeta(t, \{qx + b\}) dx = Q^{s-1} P^{t-1} C(s, t; c). \quad (3.1)$$

Both sides are jointly holomorphic in this domain and their meromorphic continuations agree. Thus arbitrary spectral derivatives follow by differentiating the displayed formula.

Proof. The disjoint endpoint singularities imply local domination by $t^{-\sigma}(1 + |\log t|^M)$ for some $\sigma < 1$ on every compact parameter set. This proves convergence and holomorphy, including all spectral jets. Initially take $\Re s, \Re t < 0$, where the Fourier series converge absolutely. A surviving frequency satisfies $pn + qm = 0$, whose complete integer solution is $n = Qr$, $m = -Pr$. For nonzero r , the Hurwitz Fourier coefficient satisfies

$$\widehat{Z}_s(Qr) = Q^{s-1} \widehat{Z}_s(r), \quad \widehat{Z}_t(-Pr) = P^{t-1} \widehat{Z}_t(-r).$$

The phase is $\exp(2\pi i r(Qa - Pb)) = \exp(-2\pi i r c)$. This is exactly the Fourier series for the right side of (3.1). Analytic continuation proves the claims. $\square$

3.2 A finite polynomial for every scale and derivative

For a periodic distribution T , let $P_{p,a}T$ be its pullback under $x \mapsto px + a$. For functions it is $T(\{px + a\})$. Put

$$\Delta_{p,a} = P_{p,a} \delta_0 = \frac{1}{p} \sum_{x_0: px_0 + a = 0 \pmod{1}} \delta_{x_0}.$$

Let G_n be the canonical periodic finite part of γ_n , characterized by

$$\langle G_n, \phi \rangle = \text{CT}_{\epsilon \downarrow 0} \int_{\epsilon}^1 \gamma_n(x) \phi(x) dx.$$

At each singular point x_0 of an affine factor use the right coordinate $t = x - x_0$. Define

$$H_{p,a;n,k} = \text{FP}_x [p^k \gamma_n^{(k)}(\{px + a\})], \quad n, k \geq 0,$$

where FP_x retains the constant term of the cutoff integral after deleting all negative powers of the local cutoffs and all nonconstant powers of their logarithms. Cutoffs at different points tend to zero independently. This specifies the coordinate t , rather than the rescaled coordinate pt .

Set

$$\begin{aligned} E_k(z) &= \prod_{j=1}^k (1 - z/j), \quad E_0(z) = 1, \\ b_{n,k}(L) &= n! [z^{n+1}] e^{Lz} E_k(z) \\ &= n! \sum_{j=0}^{\min(k, n+1)} (-1)^j e_j(k) \frac{L^{n+1-j}}{(n+1-j)!}, \end{aligned} \quad (3.2)$$

where $e_j(k)$ is the elementary symmetric polynomial of degree j in $1, 1/2, \dots, 1/k$ and $e_0(k) = 1$.

Theorem 3.2 (Dilation and differentiation contact polynomial). *For all $p \geq 1$, $n, k \geq 0$, and real shifts a ,*

$$D^k \mathbf{P}_{p,a} G_n = H_{p,a;n,k} + b_{n,k}(\log p) \Delta_{p,a}^{(k)}. \quad (3.3)$$

Consequently,

$$D^k H_{p,a;n,0} - H_{p,a;n,k} = \left[b_{n,k}(\log p) - \frac{\log^{n+1} p}{n+1} \right] \Delta_{p,a}^{(k)}. \quad (3.4)$$

In particular,

$$\mathbf{P}_{p,a} G_n = H_{p,a;n,0} + \frac{\log^{n+1} p}{n+1} \Delta_{p,a}. \quad (3.5)$$

Proof. Use the meromorphic periodic family $Z(1-z) = (\delta_0 - 1)/z + \sum_{n \geq 0} G_n z^n / n!$. Distributional differentiation and pullback commute with analytic continuation; for instance, this is first checked where all endpoint derivatives vanish and then continued. Near a singular point the k th argument derivative of the singular part is

$$(-1)^k p^{z-1} (1-z)_k t^{z-k-1}.$$

Subtract the Taylor polynomial of a test function through degree k . Only its degree- k term produces a pole at $z = 0$. In distribution notation its contribution is

$$\frac{e^{z \log p} E_k(z)}{z} \Delta_{p,a}^{(k)}.$$

The regular coefficient of this expression exceeds the fixed-coordinate constant term of the coefficient-wise cutoff integral by $[z^{n+1}] e^{z \log p} E_k(z) \Delta_{p,a}^{(k)}$. The remaining Taylor terms and the Taylor remainder are holomorphic and their coefficients are precisely the corresponding finite parts. Multiplication by $n!$ proves (3.3). Its $k = 0$ case gives (3.5); differentiating that case and subtracting the general identity proves (3.4). $\square$

The coefficients are explicit harmonic polynomials. For example,

$$b_{0,k}(L) = L - H_k, \quad b_{1,k}(L) = \frac{L^2}{2} - H_k L + \frac{H_k^2 - H_k^{(2)}}{2}.$$

At $p = 1$ this recovers the existing contact coefficient $(-1)^{n+1} n! e_{n+1}(k)$. At a nonunit scale the new logarithmic terms are essential. For example,

$$D H_{p,a;1,0} - H_{p,a;1,1} = -(\log p) \Delta'_{p,a},$$

although the same commutator vanishes at $p = 1$.

For completeness, define the transfer operator by

$$(\mathbf{L}_r f)(x) = \frac{1}{r} \sum_{j=0}^{r-1} f((x+j)/r), \quad \widehat{\mathbf{L}_r T}(k) = \widehat{T}(rk).$$

Its exact distribution relation is

$$\mathbf{L}_r G_n = \sum_{j=0}^n \binom{n}{j} (-\log r)^{n-j} G_j + \frac{(-\log r)^{n+1}}{n+1} (\delta_0 - 1). \quad (3.6)$$

Indeed, $\mathbf{L}_r Z(1-z) = r^{-z} Z(1-z)$ follows from the Fourier coefficients, and the Laurent expansion gives (3.6). The operators themselves satisfy

$$D^k \mathbf{P}_{r,0} = r^k \mathbf{P}_{r,0} D^k, \quad D^k \mathbf{L}_r = r^{-k} \mathbf{L}_r D^k, \quad \mathbf{L}_r \mathbf{P}_{r,0} = \text{Id}.$$

Thus all discrepancies arise from converting the analytic distributions to a fixed-coordinate finite part; the transfer and pullback operators have the usual exact chain rules.

3.3 A self-contained Gamma quotient for the base correlation

The reduction below is already present, in equivalent conventions, in the incoming bilinear reports. We include its short proof so that the integer-dilation identities require no external all-order closure theorem. For $0 < c < 1$, set $b = 1 - c$ and write

$$\mathcal{G}(z; c) = \zeta(1 - z, c) + \frac{1}{z} = \sum_{m \geq 0} \gamma_m(c) \frac{z^m}{m!}.$$

The singularity at $z = 0$ is removable. Define the holomorphic Gamma quotient and its removable divided difference by

$$U(z, w) = \frac{\Gamma(1 + z)\Gamma(1 - z - w)}{\Gamma(1 - w)}, \quad (3.7)$$

$$V(z, w) = \frac{U(z, w) - 1}{z(z + w)}. \quad (3.8)$$

These formulas are used near $(z, w) = (0, 0)$. Indeed, $U(0, w) = 1$ and $U(z, -z) = 1$, so the numerator is divisible by both distinct linear factors and V is holomorphic. Its exact coefficient generator is

$$U(z, w) = \exp \left(\sum_{k=2}^{\infty} \frac{\zeta(k)}{k} [(-1)^k z^k + (z + w)^k - w^k] \right). \quad (3.9)$$

In particular $V(0, 0) = \zeta(2)$, and every Taylor coefficient is a finite polynomial with rational coefficients in positive integer zeta values.

Let $J_{mn}(c)$ be the separated-endpoint, unscaled-coordinate finite part of $\gamma_m(x)\gamma_n(\{x + c\})$, and set

$$\mathcal{J}(z, w; c) = \sum_{m, n \geq 0} J_{mn}(c) \frac{z^m w^n}{m!n!}.$$

Theorem 3.3 (Holomorphic base generating identity). *With $r = z + w$, one has*

$$\begin{aligned} \mathcal{J}(z, w; c) = & \frac{\mathcal{G}(r; c) - \mathcal{G}(w; c)}{z} + \frac{\mathcal{G}(r; b) - \mathcal{G}(z; b)}{w} \\ & - V(z, w) - V(w, z) \\ & + r[V(z, w)\mathcal{G}(r; c) + V(w, z)\mathcal{G}(r; b)]. \end{aligned} \quad (3.10)$$

Every apparent singularity on the right is removable. For $M = m + n$,

$$J_{mn}(c) = \frac{\gamma_{M+1}(c)}{m+1} + \frac{\gamma_{M+1}(1-c)}{n+1} + \sum_{j=0}^{M-1} (a_j \gamma_j(c) + b_j \gamma_j(1-c)) + d, \quad (3.11)$$

where the sum is empty for $M = 0$ and the coefficients lie in $\mathbb{Q}[\zeta(2), \dots, \zeta(M+2)]$. Thus the index M is absent. The complete coefficients are obtained by multiplying $[z^m w^n]$ of (3.10) by $m!n!$.

Proof. For $\Re z, \Re w > 0$, the shifted ordinary Hurwitz integral has the meromorphic beta representation

$$C(1 - z, 1 - w; c) = B(z, 1 - r)\zeta(1 - r, c) + B(w, 1 - r)\zeta(1 - r, b). \quad (3.12)$$

Here the sum is continued before any exceptional parameter is evaluated. For completeness, the Fourier calculation first holds for $\Re z, \Re w > 1$. The positive and negative Fourier frequencies give

$$\frac{\Gamma(z)\Gamma(w)}{(2\pi)^r} \left[e^{i\pi(z-w)/2} \text{Li}_r(e^{2\pi ic}) + e^{-i\pi(z-w)/2} \text{Li}_r(e^{-2\pi ic}) \right].$$

Inserting the Hurwitz Fourier formula into the right side of (3.12), and using the Gamma reflection identity, gives this same expression. The relevant phase identity is

$$\frac{\sin(\pi w)e^{-i\pi r/2} + \sin(\pi z)e^{i\pi r/2}}{\sin(\pi r)} = e^{i\pi(z-w)/2}.$$

Local endpoint domination and analytic continuation establish (3.12) meromorphically.

The unscaled endpoint finite parts give the Laurent decomposition

$$C(1-z, 1-w; c) = -\frac{1}{zw} + \frac{\mathcal{G}(w; c)}{z} + \frac{\mathcal{G}(z; b)}{w} + \mathcal{J}(z, w; c). \quad (3.13)$$

One way to see the residue and the regular part explicitly is to use the periodic expansion $Z(1-z) = (\delta_0 - 1)/z + \sum G_m z^m/m!$. The two shifted singular supports are disjoint; their polar product has integral -1 , their mixed products evaluate the smooth opposing factor, and their regular product is precisely the prescribed local finite part. Equivalently, split the ordinary integral at $1-c$ and subtract the two leading endpoint terms before expanding. This proves (3.13) without multiplying distributions with common singular support.

Now $B(z, 1-r) = U(z, w)/z$ and $U(z, w) = 1 + zrV(z, w)$. Substitute $\zeta(1-r, c) = -1/r + \mathcal{G}(r; c)$ and its reflected analogue into (3.12). The two leading polar terms combine as $-(1/z + 1/w)/r = -1/(zw)$. Subtracting the remaining polar terms in (3.13) gives exactly (3.10).

The two divided differences contribute respectively $\gamma_{M+1}(c)/(m+1)$ and $\gamma_{M+1}(b)/(n+1)$. In the last line of (3.10), the factor r forces every Stieltjes index to be at most $M-1$. The middle line contributes only zeta-polynomial constants. Formula (3.9) then proves the coefficient ring and the missing-index assertion. $\square$

The first row is

$$J_{00}(c) = \gamma_1(c) + \gamma_1(1-c) - 2\zeta(2).$$

This checks both the Laurent signs and the finite-part constant.

3.4 All Stieltjes indices in an explicit finite formula

Define the base finite part

$$J_{mn}(c) = \text{FP} \int_0^1 \gamma_m(x) \gamma_n(\{x+c\}) dx$$

using the unscaled right coordinate at its two endpoints. Theorem 3.3 gives its all-order linear closure. For a transparent finite formula put

$$A_n(L; x) = \sum_{j=0}^n \binom{n}{j} (-L)^{n-j} \gamma_j(x),$$

$$u_n(L) = \frac{(-L)^{n+1}}{n+1}, \quad v_n(L) = \frac{L^{n+1}}{n+1}, \quad L_P = \log P, \quad L_Q = \log Q.$$

Theorem 3.4 (Unequal-dilation finite-part closure). *Under the separated-endpoint condition $0 < c < 1$, let*

$$K_{mn} = \int_{\mathbb{T}} (P_{p,a} G_m)(P_{q,b} G_n).$$

This product is well-defined because its factors have disjoint singular supports. Then

$$K_{mn} = \sum_{i=0}^m \sum_{j=0}^n \binom{m}{i} \binom{n}{j} (-L_Q)^{m-i} (-L_P)^{n-j} J_{ij}(c)$$

$$+ u_m(L_Q) A_n(L_P; c) + u_n(L_P) A_m(L_Q; 1-c)$$

$$- u_m(L_Q) u_n(L_P). \quad (3.14)$$

The finite part using the coordinate $x - x_0$ at every singular point is

$$\begin{aligned} \text{FP}_x \int_0^1 \gamma_m(\{px + a\}) \gamma_n(\{qx + b\}) dx &= K_{mn} - v_m(\log p) [A_n(L_P; c) - u_n(L_P)] \\ &\quad - v_n(\log q) [A_m(L_Q; 1 - c) - u_m(L_Q)]. \end{aligned} \quad (3.15)$$

Thus every index pair still reduces to a finite linear combination of $\gamma_j(c)$ and $\gamma_j(1 - c)$ for $0 \leq j \leq m + n + 1$, with coefficients in the ring generated over $\mathbb{Q}$ by positive integer zeta values and $\log p, \log q, \log P, \log Q$.

Proof. Write $\mathcal{G}(z; x) = \sum_{n \geq 0} \gamma_n(x) z^n / n!$ and $\mathcal{J}(z, w; c) = \sum_{m, n \geq 0} J_{mn}(c) z^m w^n / (m! n!)$. The separated-endpoint Laurent expansion is

$$C(1 - z, 1 - w; c) = -\frac{1}{zw} + \frac{\mathcal{G}(w; c)}{z} + \frac{\mathcal{G}(z; 1 - c)}{w} + \mathcal{J}(z, w; c).$$

Multiply this by $Q^{-z} P^{-w}$ as required by Theorem 3.1, and take the regular coefficient $m! n! [z^m w^n]$. The four displayed terms give the four terms in (3.14). This coefficient is the product of the regular coefficients of the two pulled-back distribution families.

To change the endpoint coordinates, apply (3.5) to the two factors. Their delta combs have disjoint supports, so the product of the two combs is zero and the two remaining correction terms evaluate the smooth opposing factor. At the first comb,

$$\frac{1}{P} \sum_{px_0 + a \in \mathbb{Z}} \gamma_n(\{qx_0 + b\}) = \frac{1}{P} \sum_{r=0}^{P-1} \gamma_n((r + c)/P) = A_n(L_P; c) - u_n(L_P).$$

The first equality uses coprimality of P, Q ; the second follows by expanding the Hurwitz multiplication formula at $s = 1$. The second comb gives $A_m(L_Q; 1 - c) - u_m(L_Q)$. This proves (3.15). The last assertion follows from the base linear closure and the explicitly triangular index sums. $\square$

At the first Stieltjes index, the existing identity $J_{00}(c) = \gamma_1(c) + \gamma_1(1 - c) - 2\zeta(2)$ gives the particularly short pulled-back answer

$$K_{00} = \gamma_1(c) + \gamma_1(1 - c) - 2\zeta(2) - L_Q \gamma_0(c) - L_P \gamma_0(1 - c) - L_P L_Q.$$

For $p = 1, q = 2$, the fixed- x version is

$$\begin{aligned} \text{FP}_x \int_0^1 \psi(x) \psi(\{2x + c\}) dx &= \gamma_1(c) + \gamma_1(1 - c) - 2\zeta(2) \\ &\quad + (\log 2)[\psi(c) + \psi(1 - c)] - \log^2 2. \end{aligned}$$

Replacing a pair (p, q) by (dp, dq) leaves the ordinary spectral integral unchanged, but in general changes its fixed-coordinate finite part. Formula (3.15) gives the complete discrepancy.

3.5 An ordinary log-Gamma identity with unequal frequencies

Let $\kappa = \gamma + \log(2\pi)$, $z_c = e^{2\pi ic}$, and let ∂_ν mean differentiation of the polylogarithm with respect to its order. The following identity needs no finite part.

Corollary 3.5 (Integer-affine log-Gamma correlation). *For the same p, q, a, b and $0 < c < 1$,*

$$\begin{aligned} \int_0^1 \log \Gamma(\{px + a\}) \log \Gamma(\{qx + b\}) dx &= \frac{\log^2(2\pi)}{4} \\ &\quad + \frac{1}{2\pi^2 PQ} \text{Re} \left[\left(\partial_\nu^2 - (2\kappa + L_P + L_Q) \partial_\nu + (\kappa + L_P)(\kappa + L_Q) + \frac{\pi^2}{4} \right) \text{Li}_\nu(z_c) \right]_{\nu=2} \\ &\quad + \frac{L_P - L_Q}{4\pi PQ} \text{Im Li}_2(z_c). \end{aligned} \quad (3.16)$$

The formula extends continuously to $c = 0$ for this ordinary integral.

Proof. Lerch's identity gives the mean-zero function $L(x) = \zeta'(0, x) = \log \Gamma(x) - \frac{1}{2} \log(2\pi)$. Its nonzero Fourier coefficients are

$$\widehat{L}(r) = \frac{\kappa + \log |r| + i\pi \operatorname{sgn}(r)/2}{2\pi i r}.$$

Either differentiate (3.1) at $(s, t) = (0, 0)$ or apply Parseval to these square-summable coefficients. Positive and negative frequencies yield

$$\begin{aligned} & \int_0^1 L(\{px + a\})L(\{qx + b\}) dx \\ &= \frac{1}{2\pi^2 PQ} \sum_{r=1}^{\infty} \frac{1}{r^2} \left[\left((\kappa + L_Q + \log r)(\kappa + L_P + \log r) + \frac{\pi^2}{4} \right) \cos(2\pi rc) \right. \\ & \quad \left. + \frac{\pi}{2} (L_P - L_Q) \sin(2\pi rc) \right]. \end{aligned}$$

The series is absolutely convergent, including two order derivatives, and converts termwise to (3.16). Restoring the mean gives the constant term. Absolute convergence also proves continuity at $c = 0$. $\square$

3.6 Audit finding and scope

The incoming archive `ProveIt_Stieltjes_Convolution_2026-10-10.zip`, `article.tex`, lines 1118–1121, suggests that changing one affine frequency produces multiple independent Fourier indices. That statement needs qualification: for two nonzero integer frequencies, the equation $pn + qm = 0$ always has the one-index parametrization used above. A suitable replacement is: “Two periodic factors with positive integer affine frequencies retain a one-dimensional frequency constraint. Their complete shifted reduction, with coordinate-dependent finite parts, is given by the integer-dilation theorem. Products of three or more independent factors generally leave a higher-dimensional frequency lattice.” This corrects an explanatory sentence in a future-work section; it does not invalidate the convolution theorems of that report.

The new formulas answer the separated-endpoint unequal-dilation problem posed in `ProveIt_Stieltjes_Correlation_Closure`, `article.tex`, lines 1247–1254, and the scale/derivative commutator problem posed in `ProveIt_Shifted_Hurwitz_Jets`, `sections.tex`, lines 782–783. They do not settle collision renormalization for two singularities at the same point or prove arithmetic independence of the resulting constant coordinates.

4 Three translated factors and colored Tornheim jets

The incoming reports ask whether three separated Stieltjes factors admit a finite coordinate description in colored Tornheim or multiple Lerch jets. We give an affirmative answer, with an explicit coefficient formula and the local subtraction convention included. The general relation between Hurwitz products and Tornheim sums is classical [6, 7, 8]; the contribution here is the separated-shift, all-Stieltjes-index formulation and its regularization.

Write $e(x) = e^{2\pi i x}$. For $|X| = |Y| = 1$ with $X, Y \neq 1$, define initially in an absolutely convergent right half-space

$$\mathcal{T}(A, B, C; X, Y) = \sum_{m, n \geq 1} \frac{X^m Y^n}{m^A n^B (m+n)^C}. \quad (4.1)$$

All powers of positive integers use their real logarithms. A superscript $\mathcal{T}^{(r, s, t)}$ below denotes derivatives in these three order variables, with X, Y fixed.

Lemma 4.1 (Entire continuation for two nontrivial twists). *For the stated twists, $\mathcal{T}$ extends to an entire function of $(A, B, C) \in \mathbb{C}^3$. For $\Re C > 0$ its continuation is*

$$\mathcal{T}(A, B, C; X, Y) = \frac{1}{\Gamma(C)} \int_0^\infty t^{C-1} \operatorname{Li}_A(Xe^{-t}) \operatorname{Li}_B(Ye^{-t}) dt. \quad (4.2)$$

The integral is locally normally convergent in A, B, C in that region. At every integer $N \geq 0$,

$$\mathcal{T}(A, B, -N; X, Y) = \sum_{j=0}^N \binom{N}{j} \text{Li}_{A-j}(X) \text{Li}_{B-N+j}(Y). \quad (4.3)$$

Proof. In an initial region, insert the Gamma integral for $(m+n)^{-C}$ and interchange absolutely convergent sums and the integral. This gives (4.2). The polylogarithm is entire in its order at each fixed argument on these two radial segments, and is holomorphic in t near zero because neither segment starts at 1. Consequently

$$f(t) = \text{Li}_A(Xe^{-t}) \text{Li}_B(Ye^{-t}) = \sum_{j=0}^M f_j(A, B) t^j + O_K(t^{M+1})$$

uniformly on compact sets K of orders. The small neighborhood in t can be chosen uniformly on K . The expansion need not converge all the way to $t = 1$; its Taylor remainder still has the asserted local estimate, which is all that subtraction requires. Thus

$$\begin{aligned} \mathcal{T}(A, B, C; X, Y) = \frac{1}{\Gamma(C)} & \left\{ \sum_{j=0}^M \frac{f_j(A, B)}{C+j} + \int_0^1 t^{C-1} \left(f(t) - \sum_{j=0}^M f_j(A, B) t^j \right) dt \right. \\ & \left. + \int_1^\infty t^{C-1} f(t) dt \right\}. \end{aligned} \quad (4.4)$$

The subtracted integral is holomorphic for $\Re C > -M-1$. The tail is entire, since each polylogarithm decays exponentially, uniformly on compact order sets. The simple zeros of $1/\Gamma(C)$ cancel the apparent poles at $C = 0, -1, \dots, -M$. Increasing M proves joint entire continuation. At $C = -N$, the value is $(-1)^N f^{(N)}(0)$. Applying $d\text{Li}_A(Xe^{-t})/dt = -\text{Li}_{A-1}(Xe^{-t})$ and Leibniz's rule gives (4.3). $\square$

4.1 A meromorphic identity before coefficient extraction

Fix $a_1, a_2, a_3 \in \mathbb{R}/\mathbb{Z}$, pairwise distinct. Fractional parts at isolated endpoints are immaterial in ordinary integrals. Put

$$K(\boldsymbol{\lambda}; \mathbf{a}) = \int_0^1 \prod_{j=1}^3 \zeta(1 - \lambda_j, \{x + a_j\}) dx, \quad \Re \lambda_j > 0. \quad (4.5)$$

For each $k \in \{1, 2, 3\}$, let $i < j$ be the other two indices and set

$$X_k = e(a_i - a_k), \quad Y_k = e(a_j - a_k), \quad \theta_k(\boldsymbol{\lambda}) = \frac{\pi}{2}(\lambda_i + \lambda_j - \lambda_k).$$

The twists are nontrivial by the separation assumption. Define

$$\begin{aligned} S(\boldsymbol{\lambda}; \mathbf{a}) = \sum_{k=1}^3 & \left\{ e^{-i\theta_k(\boldsymbol{\lambda})} \mathcal{T}(\lambda_i, \lambda_j, \lambda_k; X_k, Y_k) \right. \\ & \left. + e^{i\theta_k(\boldsymbol{\lambda})} \mathcal{T}(\lambda_i, \lambda_j, \lambda_k; X_k^{-1}, Y_k^{-1}) \right\}. \end{aligned} \quad (4.6)$$

Theorem 4.2 (Six-cone identity). *For $\Re \lambda_j > 0$,*

$$K(\boldsymbol{\lambda}; \mathbf{a}) = \frac{\prod_{j=1}^3 \Gamma(\lambda_j)}{(2\pi)^{\lambda_1 + \lambda_2 + \lambda_3}} S(\boldsymbol{\lambda}; \mathbf{a}). \quad (4.7)$$

The right side gives a meromorphic continuation to $\mathbb{C}^3$.

Proof. First assume $\Re \lambda_j > 1$. The nonzero Fourier coefficients of the periodic Hurwitz function are

$$\zeta(\widehat{1 - \lambda, \cdot})(n) = \frac{\Gamma(\lambda)}{(2\pi|n|)^\lambda} e^{-i\pi\lambda \operatorname{sgn}(n)/2}, \quad n \neq 0, \quad (4.8)$$

and the zero coefficient is zero [3, Eq. 25.11.9]. The three Fourier series converge absolutely, so integration keeps exactly $n_1 + n_2 + n_3 = 0$ with every $n_j \neq 0$. Every such triple has either two positive frequencies and one negative frequency, or the opposite sign pattern. In the first case the negative index is a unique k and the frequencies are $m, n, -m - n$. Their phase and denominator give the first term of the k th summand in (4.6). Reversing all signs gives its second term. This is a disjoint decomposition, so no factor of two or ordering multiplicity is missing.

The integral in (4.5) is jointly holomorphic on $\Re \lambda_j > 0$. Near a singular point of one factor, the other two factors are smooth and uniformly bounded on compact order sets. The local majorants are $t^{\sigma-1}(1 + |\log t|^M)$ with $\sigma > 0$. The right side is also holomorphic there by Lemma 4.1. The identity theorem extends the equality from the smaller right half-space to the claimed domain. The same formula supplies its meromorphic continuation. $\square$

4.2 The precise endpoint finite part

For $m \geq 0$, use

$$\gamma_m(t) = \frac{\log^m t}{t} + \gamma_m(1+t), \quad 0 < t < 1. \quad (4.9)$$

Let $x_k = \{-a_k\}$ and remove a right-hand arc of length ε from each x_k , using the ordinary unit coordinate on the circle. For sufficiently small ε these arcs are disjoint. Denote the remaining set by E_ε , and put

$$A_k(\mathbf{m}; \mathbf{a}) = \prod_{j \neq k} \gamma_{m_j}(\{a_j - a_k\}).$$

Definition 4.3 (Separated triple finite part). For $m_1, m_2, m_3 \geq 0$, define

$$I_{\mathbf{m}}(\mathbf{a}) = \lim_{\varepsilon \downarrow 0} \left\{ \int_{E_\varepsilon} \prod_{j=1}^3 \gamma_{m_j}(\{x + a_j\}) dx + \sum_{k=1}^3 A_k(\mathbf{m}; \mathbf{a}) \frac{\log^{m_k+1} \varepsilon}{m_k + 1} \right\}. \quad (4.10)$$

The limit exists: after subtracting its displayed leading singular term, the local product is $O(1 + |\log t|^{m_k})$, hence integrable. No product of distributions with common singular support is used.

Theorem 4.4 (All-index triple Stieltjes identity). *Let*

$$Q(\boldsymbol{\lambda}; \mathbf{a}) = \prod_{j=1}^3 \left\{ \Gamma(1 + \lambda_j) (2\pi)^{-\lambda_j} \right\} S(\boldsymbol{\lambda}; \mathbf{a}). \quad (4.11)$$

This function is holomorphic near $\boldsymbol{\lambda} = 0$, and

$$I_{\mathbf{m}}(\mathbf{a}) = \left(\prod_{j=1}^3 m_j! \right) [\lambda_1^{m_1+1} \lambda_2^{m_2+1} \lambda_3^{m_3+1}] Q(\boldsymbol{\lambda}; \mathbf{a}). \quad (4.12)$$

Thus the answer uses finitely many derivatives of six colored Tornheim kernels, of total order at most $m_1 + m_2 + m_3 + 3$.

Proof. In a local right coordinate t at x_k , split the k th Hurwitz factor as $t^{\lambda_k-1} + \zeta(1 - \lambda_k, 1 + t)$. For a smooth local multiplier g , analytic continuation of the singular term is obtained from

$$\int_0^h t^{\lambda-1} g(t) dt = \frac{g(0)h^\lambda}{\lambda} + \int_0^h t^{\lambda-1} (g(t) - g(0)) dt.$$

The constant and higher Laurent coefficients of this formula are the Hadamard finite parts of $\log^m(t)g(t)/t$, divided by $m!$. Use disjoint neighborhoods of the three singularities and an ordinary integral on their complement. The other factors are smooth in the local coordinate, although meromorphic in their own order variables. Here is a uniform justification for extracting all three coefficients. Let $g_k(t)$ be the product of those other factors, and define

$$F_\varepsilon(\lambda) = \int_{E_\varepsilon} \prod_j \zeta(1 - \lambda_j, \{x + a_j\}) dx + \sum_k \frac{\varepsilon^{\lambda_k} g_k(0)}{\lambda_k}.$$

On a small closed polydisk $|\lambda_j| \leq \rho < 1$, with $i, j \neq k$, one has $\lambda_i \lambda_j (g_k(t) - g_k(0)) = O(t)$ and $\lambda_k \zeta(1 - \lambda_k, 1 + t) = O(1)$ uniformly. The local decomposition therefore gives

$$\lambda_1 \lambda_2 \lambda_3 (K - F_\varepsilon) = O(\varepsilon^{1-\rho})$$

uniformly on that polydisk. After the indicated pole factors are removed, all these expressions are holomorphic. Cauchy's coefficient formula permits passage to the limit. The counterterms contribute exactly those in (4.10), and hence

$$[\lambda_1^{m_1} \lambda_2^{m_2} \lambda_3^{m_3}] K = \frac{I_m}{m_1! m_2! m_3!}.$$

The convention $\zeta(1 - \lambda, t) = -1/\lambda + \sum_{m \geq 0} \gamma_m(t) \lambda^m / m!$ explains why this equation has no extra alternating signs. Finally $K = Q/(\lambda_1 \lambda_2 \lambda_3)$ by $\Gamma(1 + \lambda) = \lambda \Gamma(\lambda)$, proving (4.12). $\square$

4.3 An explicit formula for three digamma factors

Set $L = \gamma + \log(2\pi)$, where $\gamma = \gamma_0(1)$, and introduce

$$D_+ = \partial - L - \frac{i\pi}{2}, \quad D_- = \partial - L + \frac{i\pi}{2}.$$

The variable differentiated by each operator is specified by a subscript. Since $\Gamma(1 + \lambda) = 1 - \gamma\lambda + O(\lambda^2)$, Theorem 4.4 immediately gives

$$\boxed{\text{FP} \int_0^1 \prod_{j=1}^3 \psi(\{x + a_j\}) dx = -2\Re \sum_{k=1}^3 D_{+,A} D_{+,B} D_{-,C} \mathcal{T}(A, B, C; X_k, Y_k)|_{A=B=C=0}.} \quad (4.13)$$

The finite part on the left is the negative of $I_{(0,0,0)}$, because $\gamma_0 = -\psi$. This settles the specific three-digamma coordinate question in the shifted-jets report. It does not assert that these double-zeta derivatives reduce to ordinary single zeta or Stieltjes values.

For example, for shifts $0, 1/3, 2/3$, write $\omega = e(1/3)$. Symmetry of $\mathcal{T}$ in (A, X) and (B, Y) shows that all three terms agree:

$$\text{FP} \int_0^1 \psi(x) \psi(\{x + \frac{1}{3}\}) \psi(\{x + \frac{2}{3}\}) dx = -6\Re D_{+,A} D_{+,B} D_{-,C} \mathcal{T}(A, B, C; \omega, \omega^2)|_{A=B=C=0}. \quad (4.14)$$

This is an exact root-of-unity specialization with a fully specified regularization, not a numerical relation guess.

4.4 A convergent representation of the first spectral derivative

The first derivative in the final Tornheim order is especially convenient for independent numerical checking. With $f(t) = \text{Li}_A(Xe^{-t}) \text{Li}_B(Ye^{-t})$, equation (4.4) at $M = 0$ gives

$$\partial_C \mathcal{T}(A, B, C; X, Y)|_{C=0} = \gamma f(0) + \int_0^1 \frac{f(t) - f(0)}{t} dt + \int_1^\infty \frac{f(t)}{t} dt. \quad (4.15)$$

All A, B derivatives pass through these convergent integrals on compact sets. Let

$$F_X(t) = \partial_s \text{Li}_s(Xe^{-t}) \Big|_{s=0} - (L + i\pi/2) \text{Li}_0(Xe^{-t}), \quad h(t) = F_X(t) F_Y(t).$$

Then the quantity in one summand of (4.13) is

$$\int_0^1 \frac{h(t) - h(0)}{t} dt + \int_1^\infty \frac{h(t)}{t} dt + \left(-\log(2\pi) + \frac{i\pi}{2} \right) h(0). \quad (4.16)$$

The Euler constant in the last coefficient has canceled against the $\gamma f(0)$ term in (4.15). Dropping that term would give an incorrect finite constant.

4.5 Integral inner orders and multiple polylogarithms

For positive integers A, B and initially sufficiently large $\Re C$, partial fractions give the classical colored Tornheim reduction [7]:

$$\begin{aligned} \mathcal{T}(A, B, C; X, Y) &= \sum_{j=0}^{A-1} \binom{B+j-1}{j} \text{Li}_{B+C+j, A-j}(Y, X/Y) \\ &\quad + \sum_{j=0}^{B-1} \binom{A+j-1}{j} \text{Li}_{A+C+j, B-j}(X, Y/X), \end{aligned} \quad (4.17)$$

where $\text{Li}_{p,q}(z, w) = \sum_{k>m \geq 1} z^k w^m / (k^p m^q)$. For completeness, multiply the elementary identity

$$\frac{1}{m^A n^B} = \sum_{j=0}^{A-1} \binom{B+j-1}{j} \frac{1}{m^{A-j} (m+n)^{B+j}} + \sum_{j=0}^{B-1} \binom{A+j-1}{j} \frac{1}{n^{B-j} (m+n)^{A+j}}$$

by $X^m Y^n (m+n)^{-C}$ and substitute $k = m+n$. The displayed elementary identity follows inductively from $1/(mn) = (1/m + 1/n)/(m+n)$. Analytic continuation then extends the complete expression in C . At rational shifts, all the colors in (4.17) are roots of unity. One may differentiate this identity in C . One must not differentiate it in the discrete positive-integer parameters A, B ; it provides no such identity for those derivatives at zero.

4.6 Ordinary triple log-Gamma correlations

Let $G(x) = \log \Gamma(x) - \frac{1}{2} \log(2\pi) = \zeta'(0, x)$. Each translated G has only a logarithmic singularity, so its triple product is integrable. Since $s_j = 1 - \lambda_j$, differentiating (4.5) once in each s_j at zero proves the ordinary identity

$$\int_0^1 \prod_{j=1}^3 G(\{x + a_j\}) dx = -\partial_{\lambda_1} \partial_{\lambda_2} \partial_{\lambda_3} \left\{ \frac{\prod_j \Gamma(\lambda_j)}{(2\pi)^{\sum_j \lambda_j}} S(\boldsymbol{\lambda}; \mathbf{a}) \right\} \Big|_{\lambda_1=\lambda_2=\lambda_3=1}. \quad (4.18)$$

Here no finite part is present. More generally, any fixed spectral derivatives at $s_j = -N_j \leq 0$ give convergent triple integrals of Hurwitz jets, which include the normalized Stieltjes antiderivatives. The local estimates are products of powers of $\log t$ times nonnegative powers of t , so differentiation under

the integral is justified. This extends the same six-coordinate mechanism from singular Stieltjes functions to their ordinary integrated relatives.

Here is an explicit version at every primitive order. For $r \geq 1$ set

$$V_{m,r}(x) = (-1)^{m+1} m! [u^{m+1}] \frac{\zeta(1-r+u, x)}{(r-1)! \prod_{j=1}^{r-1} (1-u/j)}. \quad (4.19)$$

Empty products equal one. These are finite linear combinations of Hurwitz order derivatives at $1-r$ and have mean zero on the unit interval. In particular $V_{0,1} = -G$.

Corollary 4.5 (All triple Stieltjes primitive correlations). *The pointwise identity $\partial_x^r V_{m,r}(x) = \gamma_m(x)$ holds on $0 < x < 1$. For pairwise distinct shifts and integers $r_j \geq 1, m_j \geq 0$,*

$$\begin{aligned} & \int_0^1 \prod_{j=1}^3 V_{m_j, r_j}(\{x + a_j\}) dx \\ &= (-1)^{m_1+m_2+m_3+3} \prod_{j=1}^3 m_j! [u_1^{m_1+1} u_2^{m_2+1} u_3^{m_3+1}] \frac{K(\mathbf{r} - \mathbf{u}; \mathbf{a})}{\prod_{j=1}^3 \{(r_j-1)! \prod_{h=1}^{r_j-1} (1-u_j/h)\}}. \end{aligned} \quad (4.20)$$

In this formula K is explicitly the six-cone right side of (4.7), and the integral is ordinary and absolutely convergent.

Proof. The repeated Hurwitz derivative law gives

$$\partial_x^r \zeta(1-r+u, x) = -u(r-1)! \prod_{j=1}^{r-1} (1-u/j) \zeta(1+u, x).$$

Extract the coefficient in (4.19) and use the defining Stieltjes Laurent expansion. This proves the primitive identity, including its sign and normalization. The identity $\int_0^1 \zeta(1-r+u, x) dx = 0$ for small u proves the asserted mean-zero normalization. Near $x = 0$, write the zeta function as $x^{r-1-u} + \zeta(1-r+u, 1+x)$. On $|u| \leq \rho < 1$, its derivatives of any fixed order are bounded by a constant times $1 + x^{r-1-\rho}(1 + |\log x|^M)$ for some finite M . This is integrable even at $r = 1$. At each singularity of the product, the other two shifted factors are smooth; the majorant is therefore uniform on a sufficiently small polydisk of the u_j . One can extract all coefficients under the integral, obtaining (4.20). $\square$

5 A Gamma coefficient calculus for shifted harmonic sums

The exact families in this section complement the manuscript's Hurwitz-jet calculus. They involve a nonlinear but structured class of generalized harmonic-number numerators and admit a finite reduction for every rational denominator that decays quadratically. The unshifted specialization is classical: it recovers the Aomoto–Drinfeld–Zagier generating function for height-one multiple zeta values. Young's fractional-order Barnes–Hurwitz generating function [9], and the height-one regularization formulas recalled by Kaneko and Sakata [11], are directly relevant prior work. Our purpose is a proved shifted-parameter calculus, with a precise two-variable explanation of its finite parts. We do not claim that the kernel, its unshifted specialization, or its classical low-depth Euler sums are new.

Throughout this section, $\Re a > 0$ and $\log(n+a)$ is the principal logarithm. Define the elementary harmonic coefficients by

$$\prod_{k=1}^n \left(1 + \frac{y}{k}\right) = \sum_{r=0}^n E_{n,r} y^r, \quad E_{0,0} = 1, \quad E_{n,r} = 0 \quad (r > n). \quad (5.1)$$

Thus, with $H_n^{(j)} = \sum_{k=1}^n k^{-j}$ and $H_n = H_n^{(1)}$,

$$E_{n,0} = 1, \quad E_{n,1} = H_n, \quad (5.2)$$

$$E_{n,2} = \frac{H_n^2 - H_n^{(2)}}{2}, \quad E_{n,3} = \frac{H_n^3 - 3H_n H_n^{(2)} + 2H_n^{(3)}}{6}. \quad (5.3)$$

Equivalently, $E_{n,r}$ is the sum of $1/(k_1 \cdots k_r)$ over $1 \leq k_1 < \cdots < k_r \leq n$. In particular, $E_{n,r} = O_r((1 + \log(n+1))^r)$.

The Gamma coefficients to which these sums reduce are

$$F(a, y) := \frac{\Gamma(a)\Gamma(1-y)}{\Gamma(a-y)} = \sum_{k=0}^{\infty} C_k(a) y^k. \quad (5.4)$$

This is analytic for sufficiently small y , and $C_0(a) = 1$. Put

$$A_1(a) = \gamma + \psi(a), \quad A_j(a) = \zeta(j) - \zeta(j, a) \quad (j \geq 2). \quad (5.5)$$

The Gamma Taylor series gives

$$F(a, y) = \exp \left(\sum_{j=1}^{\infty} \frac{A_j(a)}{j} y^j \right), \quad k C_k(a) = \sum_{j=1}^k A_j(a) C_{k-j}(a). \quad (5.6)$$

Consequently every C_k is a polynomial with rational coefficients in generalized harmonic functions. For a positive integer $a = N$, $A_j(N) = H_{N-1}^{(j)}$, and $C_k(N)$ is the complete homogeneous harmonic sum of depth k on $1, \dots, N-1$.

For compact formulas, write

$$A = \gamma + \psi(a), \quad B = \zeta(2) - \psi'(a), \quad C = \zeta(3) + \frac{\psi''(a)}{2}, \quad D = \zeta(4) - \frac{\psi^{(3)}(a)}{6}. \quad (5.7)$$

Then

$$C_1(a) = A, \quad (5.8)$$

$$C_2(a) = \frac{A^2 + B}{2}, \quad (5.9)$$

$$C_3(a) = \frac{A^3 + 3AB + 2C}{6}, \quad (5.10)$$

$$C_4(a) = \frac{A^4 + 6A^2B + 3B^2 + 8AC + 6D}{24}. \quad (5.11)$$

5.1 An exact generating theorem

Define

$$Z_r(s, a) := \sum_{n=0}^{\infty} \frac{E_{n,r}}{(n+a)^s}, \quad \Re s > 1. \quad (5.12)$$

For $a = 1$, the convention with the largest index first gives $Z_r(s, 1) = \zeta(s, \{1\}^r)$.

Theorem 5.1 (Shifted harmonic Gamma generating function). *For $|x| < \Re a$ and $|y| < 1$,*

$$\sum_{m=1}^{\infty} \sum_{r=0}^{\infty} Z_r(m+1, a) x^m y^{r+1} = F(a, y) - F(a-x, y). \quad (5.13)$$

In particular, for all integers $r \geq 0$ and $m \geq 1$,

$$\boxed{Z_r(m+1, a) = \frac{(-1)^{m+1}}{m!} \frac{d^m}{da^m} C_{r+1}(a).} \quad (5.14)$$

For $\Re a, \Re b > 0$, the endpoint difference is the convergent identity

$$\boxed{\sum_{n=0}^{\infty} E_{n,r} \left(\frac{1}{n+a} - \frac{1}{n+b} \right) = C_{r+1}(b) - C_{r+1}(a).} \quad (5.15)$$

Proof. The ordinary generating function of the elementary harmonic coefficients follows from the binomial theorem:

$$\sum_{n=0}^{\infty} E_{n,r} u^n = \frac{(-\log(1-u))^r}{r!(1-u)}, \quad |u| < 1. \quad (5.16)$$

Indeed, $\sum_n (1+y)_n u^n / n! = (1-u)^{-1-y}$, and extraction of y^r gives the assertion. The integrable difference in (5.15) therefore has generating function

$$\begin{aligned} & \sum_{r \geq 0} y^r \sum_{n \geq 0} E_{n,r} \left(\frac{1}{n+a} - \frac{1}{n+b} \right) \\ &= \int_0^1 (u^{a-1} - u^{b-1}) (1-u)^{-1-y} du = B(a, -y) - B(b, -y) = \frac{F(b, y) - F(a, y)}{y}. \end{aligned} \quad (5.17)$$

The difference integral converges for $\Re y < 1$: its numerator vanishes to first order at $u = 1$. Equality with the difference of beta functions first holds for $\Re y < 0$ and extends by holomorphy, with the singularity at $y = 0$ removable. Near $u = 0$, $\Re a, \Re b > 0$ provides an integrable majorant. For the sums, absolute and locally uniform convergence follow from $\sum_r E_{n,r} |y|^r = (1+|y|)_n / n! = O(n^{|y|})$ and the quadratic decay of the resolvent difference. Coefficient extraction proves (5.15).

Apply the same identity with $b = a$ and the first parameter $a - x$. Since

$$\frac{1}{n+a-x} - \frac{1}{n+a} = \sum_{m=1}^{\infty} \frac{x^m}{(n+a)^{m+1}},$$

the claimed double generating function follows. The imposed bound on x justifies this expansion uniformly in n , and coefficient extraction from $F(a-x, y)$ gives (5.14). $\square$

At $a = 1$, $F(1, y) = 1$, so (5.13) becomes

$$\sum_{m \geq 1, r \geq 0} \zeta(m+1, \{1\}^r) x^m y^{r+1} = 1 - \frac{\Gamma(1-x)\Gamma(1-y)}{\Gamma(1-x-y)},$$

the classical height-one generating function. In this specialization the right side is symmetric in x, y , giving the standard height-one duality. There is generally no such symmetry for an arbitrary shift a .

5.2 Explicit polygamma reductions and antiderivatives

Set $P_j = \psi^{(j)}(a)$, and retain $C_0 = 1$ and the explicit C_k in (5.11). The first two denominator powers, at depths zero through three, are as follows:

$$Z_0(2, a) = P_1, \quad (5.18)$$

$$Z_1(2, a) = C_1 P_1 - \frac{P_2}{2}, \quad (5.19)$$

$$Z_2(2, a) = C_2 P_1 - \frac{C_1 P_2}{2} + \frac{P_3}{6}, \quad (5.20)$$

$$Z_3(2, a) = C_3 P_1 - \frac{C_2 P_2}{2} + \frac{C_1 P_3}{6} - \frac{P_4}{24}; \quad (5.21)$$

and

$$Z_0(3, a) = -\frac{P_2}{2}, \quad (5.22)$$

$$Z_1(3, a) = -\frac{1}{2} \left(C_1 P_2 + P_1^2 - \frac{P_3}{2} \right), \quad (5.23)$$

$$Z_2(3, a) = -\frac{1}{2} \left[C_2 P_2 + C_1 \left(P_1^2 - \frac{P_3}{2} \right) + \frac{P_4}{6} - P_1 P_2 \right], \quad (5.24)$$

$$Z_3(3, a) = -\frac{1}{2} \left[C_3 P_2 + C_2 \left(P_1^2 - \frac{P_3}{2} \right) + C_1 \left(\frac{P_4}{6} - P_1 P_2 \right) - \frac{P_5}{24} + \frac{P_1 P_3}{3} + \frac{P_2^2}{4} \right]. \quad (5.25)$$

These are finite closed forms, not recursively defined Euler sums. For example,

$$\sum_{n=0}^{\infty} \frac{H_n}{(n+a)^2} = (\psi(a) + \gamma) \psi'(a) - \frac{\psi''(a)}{2}. \quad (5.26)$$

Each entry is an explicit antiderivative formula as well:

$$\int Z_r(2, a) da = C_{r+1}(a) + \text{constant}, \quad \int Z_r(m+1, a) da = -\frac{1}{m} Z_r(m, a) + \text{constant} \quad (m \geq 2). \quad (5.27)$$

The first formula identifies a finite polynomial in polygammas as the primitive; it avoids leaving an unevaluated harmonic Dirichlet series.

For a useful nonintegral specialization, put $L = \log 2$. The half-point values $\zeta(j, 1/2) = (2^j - 1)\zeta(j)$ give

$$Z_1(2, 1/2) = 7\zeta(3) - 6L\zeta(2), \quad (5.28)$$

$$Z_2(2, 1/2) = \frac{15}{2}\zeta(4) - 14L\zeta(3) + 6L^2\zeta(2), \quad (5.29)$$

$$Z_3(2, 1/2) = 31\zeta(5) - 13\zeta(2)\zeta(3) - 15L\zeta(4) + 14L^2\zeta(3) - 4L^3\zeta(2). \quad (5.30)$$

Thus the last identity, written without $E_{n,3}$, is

$$\begin{aligned} & \sum_{n=0}^{\infty} \frac{H_n^3 - 3H_n H_n^{(2)} + 2H_n^{(3)}}{(2n+1)^2} \\ &= \frac{3}{2} \left[31\zeta(5) - 13\zeta(2)\zeta(3) - 15L\zeta(4) + 14L^2\zeta(3) - 4L^3\zeta(2) \right]. \end{aligned} \quad (5.31)$$

The factor $3/2$ is $3!/4$, from the harmonic normalization and the conversion $(n+1/2)^{-2} = 4(2n+1)^{-2}$.

5.3 Closure under arbitrary rational kernels

Corollary 5.2 (Rational-kernel reduction). *Let $R(z)$ be rational, have all its finite poles among $-a_1, \dots, -a_J$ with $\Re a_j > 0$, and satisfy $R(z) = O(z^{-2})$ at infinity. Write*

$$R(z) = \sum_{j=1}^J \sum_{\ell=1}^{L_j} \frac{c_{j,\ell}}{(z+a_j)^\ell}.$$

Then $\sum_j c_{j,1} = 0$, and

$$\begin{aligned} \sum_{n=0}^{\infty} E_{n,r} R(n) &= - \sum_{j=1}^J c_{j,1} C_{r+1}(a_j) \\ &\quad + \sum_{j=1}^J \sum_{\ell=2}^{L_j} \frac{(-1)^\ell c_{j,\ell}}{(\ell-1)!} C_{r+1}^{(\ell-1)}(a_j). \end{aligned} \quad (5.32)$$

Proof. The vanishing sum of residues is the coefficient of z^{-1} at infinity. It permits replacing each simple-pole summand by its difference from a fixed reference resolvent. Apply (5.15) to these differences and (5.14) to all repeated poles. All combined series are absolutely convergent. $\square$

For real $a > 0$ and real $b \neq 0$, a compact example is

$$\sum_{n=0}^{\infty} \frac{E_{n,r}}{(n+a)^2 + b^2} = \frac{C_{r+1}(a+ib) - C_{r+1}(a-ib)}{2ib} = \frac{\Im C_{r+1}(a+ib)}{b}. \quad (5.33)$$

The limit $b \rightarrow 0$ recovers $Z_r(2, a) = C'_{r+1}(a)$.

5.4 The moving pole and all-depth finite parts

The noninteger order of the Barnes–Hurwitz generating kernel is the right variable for explaining why the endpoint finite parts are also Gamma coefficients. This is more delicate than substituting $s = 1$ into the divergent defining series.

Theorem 5.3 (Two-variable pole removal). *For $\Re a > 0$, initially define*

$$\mathcal{B}(s, y; a) = \sum_{n=0}^{\infty} \frac{(1+y)_n}{n!(n+a)^s}, \quad \Re s > 1 + \Re y. \quad (5.34)$$

Near $(s, y) = (1, 0)$ there is a jointly holomorphic function $\mathcal{R}(s, y; a)$ such that

$$\Gamma(s)\mathcal{B}(s, y; a) = \frac{1}{s-1-y} + \mathcal{R}(s, y; a), \quad \mathcal{R}(1, y; a) = \frac{1 - F(a, y)}{y}. \quad (5.35)$$

If

$$\frac{1}{\Gamma(1+z)} = \sum_{j=0}^{\infty} g_j z^j, \quad (5.36)$$

then

$$Z_r(1+z, a) = \sum_{j=0}^r \frac{g_j}{z^{r+1-j}} + g_{r+1} - C_{r+1}(a) + O(z). \quad (5.37)$$

The $O(z)$ term is locally uniform in a on the right half-plane.

Proof. The Mellin representation, initially in a common half-plane where absolute convergence justifies summation under the integral, is

$$\Gamma(s)\mathcal{B}(s, y; a) = \int_0^{\infty} t^{s-1} e^{-at} (1 - e^{-t})^{-1-y} dt.$$

Split it at $t = 1$ and remove its leading small- t term. The remainder is

$$\begin{aligned} \mathcal{R}(s, y; a) &= \int_0^1 t^{s-2-y} \left[e^{-at} \left(\frac{t}{1-e^{-t}} \right)^{1+y} - 1 \right] dt \\ &\quad + \int_1^{\infty} t^{s-1} e^{-at} (1 - e^{-t})^{-1-y} dt. \end{aligned} \quad (5.38)$$

The bracket is $O(t)$ uniformly on compact subsets of (y, a) with $\Re a > 0$. Hence the first integral is holomorphic whenever $\Re(s-y) > 0$, and the second is holomorphic in s, y because of exponential decay. Derivatives introduce only bounded powers of $\log t$ near zero and polynomial factors at infinity; the same majorants justify all mixed derivatives on compact subsets. The subtracted integral is $(s-1-y)^{-1}$.

For $\Re y < 0$, substitute $u = e^{-t}$ at $s = 1$ to obtain

$$\mathcal{B}(1, y; a) = B(a, -y) = -\frac{F(a, y)}{y}.$$

Consequently $\mathcal{R}(1, y; a) = (1 - F(a, y))/y$, and this equality extends holomorphically across $y = 0$.

Extracting y^r first in a region $|y| < |s - 1|$ gives

$$\Gamma(s)Z_r(s, a) = \frac{1}{(s-1)^{r+1}} + R_r(s, a), \quad R_r(s, a) := [y^r]\mathcal{R}(s, y; a), \quad R_r(1, a) = -C_{r+1}(a).$$

Both sides extend meromorphically to a neighborhood of $s = 1$. Multiply by the reciprocal Gamma Taylor series. Its coefficient g_{r+1} from the pole contributes to the constant term, in addition to $R_r(1, a)$. This proves (5.37). $\square$

The first reciprocal Gamma coefficients are

$$g_0 = 1, \tag{5.39}$$

$$g_1 = \gamma, \tag{5.40}$$

$$g_2 = \frac{\gamma^2 - \zeta(2)}{2}, \tag{5.41}$$

$$g_3 = \frac{\gamma^3 - 3\gamma\zeta(2) + 2\zeta(3)}{6}, \tag{5.42}$$

$$g_4 = \frac{\gamma^4 - 6\gamma^2\zeta(2) + 3\zeta(2)^2 + 8\gamma\zeta(3) - 6\zeta(4)}{24}. \tag{5.43}$$

Thus the complete principal parts and constants for $r = 0, 1, 2, 3$ are

$$Z_0(1 + z, a) = z^{-1} + \gamma - A + O(z), \tag{5.44}$$

$$Z_1(1 + z, a) = z^{-2} + \gamma z^{-1} + g_2 - \frac{A^2 + B}{2} + O(z), \tag{5.45}$$

$$Z_2(1 + z, a) = z^{-3} + \gamma z^{-2} + g_2 z^{-1} + g_3 - \frac{A^3 + 3AB + 2C}{6} + O(z), \tag{5.46}$$

$$Z_3(1 + z, a) = z^{-4} + \gamma z^{-3} + g_2 z^{-2} + g_3 z^{-1} + g_4 - \frac{A^4 + 6A^2B + 3B^2 + 8AC + 6D}{24} + O(z). \tag{5.47}$$

For $a = 1$, $C_k(1) = 0$ for every $k \geq 1$, and the constant is simply g_{r+1} , consistently with height-one multiple-zeta regularization.

5.5 Stieltjes constants, the literature comparison, and limits of the reduction

Use the conventional normalization

$$\zeta(1 + z, a) = \frac{1}{z} + \sum_{m=0}^{\infty} \frac{(-1)^m \gamma_m(a)}{m!} z^m. \tag{5.48}$$

The $r = 0$ finite-part formula says $\gamma_0(a) = \gamma - C_1(a) = -\psi(a)$, as required. More generally let

$$\Gamma(1 + z) = \sum_{j \geq 0} q_j z^j, \quad R_{r,k}(a) = [y^r z^k] \mathcal{R}(1 + z, y; a).$$

The ordinary Stieltjes layer is exactly

$$R_{0,k}(a) = q_{k+1} + \sum_{j=0}^k q_{k-j} \frac{(-1)^j \gamma_j(a)}{j!}. \tag{5.49}$$

For every $r \geq 0$ the coefficient of z^m , $m \geq 0$, in the full Laurent expansion of $Z_r(1+z, a)$ is

$$d_{r,m}(a) = g_{r+m+1} + \sum_{k=0}^m g_{m-k} R_{r,k}(a). \quad (5.50)$$

Our finite-part theorem determines $R_{r,0}(a) = -C_{r+1}(a)$. It does *not* determine $R_{r,k}$ for $r, k \geq 1$ from Gamma coefficients. Those mixed Taylor coefficients are higher harmonic Stieltjes data. Any claim that all of them reduce to polygamma values would require additional identities.

The distinction between two shifted conventions matters. Kargin, Dil, Cenkci, and Can [12] study

$$\zeta_H(s, a) = \sum_{n=0}^{\infty} \frac{\sum_{k=0}^n (k+a)^{-1}}{(n+a)^s}.$$

For general a , its harmonic numerator is shifted along with its denominator, whereas ours is the unshifted H_n . The two should not be identified. At $a = 1$ they satisfy

$$Z_1(s, 1) = \zeta_H(s, 1) - \zeta(s+1). \quad (5.51)$$

The known constant term $\gamma_H(0, 1) = (\gamma^2 + \zeta(2))/2$, quoted in [12], therefore gives $\text{FP}_{s=1} Z_1(s, 1) = (\gamma^2 - \zeta(2))/2 = g_2$, exactly as above. Their higher harmonic Stieltjes constants, and Young's higher-depth analogues [10, 9], are the appropriate comparison for the still nontrivial mixed coefficients. Coffey's nonlinear harmonic reductions [13] already include familiar specializations such as $Z_1(2, 1) = \zeta(3)$ and $Z_2(2, 1) = \zeta(4)$.

An entirely convergent integral representation of the mixed data is available directly from (5.38). Put

$$K_r(t, a) = \frac{e^{-at} [-\log(1 - e^{-t})]^r}{r!(1 - e^{-t})}.$$

Then

$$R_{r,k}(a) = \frac{1}{k!} \left\{ \int_0^1 (\log t)^k \left[K_r(t, a) - \frac{(-\log t)^r}{r!t} \right] dt + \int_1^{\infty} (\log t)^k K_r(t, a) dt \right\}. \quad (5.52)$$

For example, $R_{0,1}(a)$ combines with $R_{0,0}(a)$ by (5.50) to give $-\gamma_1(a)$. Formula (5.52) is a useful exact definition and a numerical verification route, but is not asserted to be a closed evaluation of all mixed data.

6 Gauss summation at zero balance and Stieltjes jets

The circular correlation identities in the incoming manuscripts concern products of periodic Hurwitz functions. The present section develops a different, discrete construction: a gamma-ratio coefficient is compared term by term with its leading Hurwitz-zeta coefficient. At zero balance, this subtraction converts the pole in Gauss summation into an ordinary, absolutely convergent series. Its complete parameter jet contains generalized Stieltjes constants, polygamma functions, and nonlinear generalized harmonic sums.

Gauss summation, gamma-ratio asymptotics, and the use of hypergeometric parameter derivatives to evaluate harmonic sums are classical [14, 15, 16]. Hypergeometric representations of Stieltjes constants also have an established literature [17]. The contribution here is an explicit common normalization at zero balance, an every-index mixed-jet statement, its polylogarithmic boundary interpretation, and concrete corrected formulas for a group of historically printed conjectures. No claim of first discovery of Gauss differentiation, or of the present-day open status of those conjectures, is made.

6.1 A convergent master identity

Use the convention

$$\zeta(1+s, c) = \frac{1}{s} + \sum_{k=0}^{\infty} \frac{(-1)^k \gamma_k(c)}{k!} s^k. \quad (6.1)$$

Initially $a, b, c > 0$. All statements below extend holomorphically to $\Re a, \Re b, \Re c > 0$, with the principal logarithms of $n+c$. Define

$$W_n(a, b; s) = \frac{\Gamma(n+a)\Gamma(n+b)}{\Gamma(n+a+b+s)\Gamma(n+1)}, \quad (6.2)$$

$$R_{a,b}(s) = \frac{\Gamma(1+s)\Gamma(a)\Gamma(b)}{\Gamma(a+s)\Gamma(b+s)}. \quad (6.3)$$

Reciprocal gamma factors are interpreted as entire functions; a zero of such a factor is not an excluded parameter value.

Theorem 6.1 (Gauss–Stieltjes subtraction). *For $\Re a, \Re b, \Re c > 0$ and $\Re s > -1$, the series*

$$\mathcal{F}_{a,b,c}(s) = \sum_{n=0}^{\infty} \left(W_n(a, b; s) - (n+c)^{-1-s} \right) \quad (6.4)$$

converges normally on compact subsets of the full parameter domain and is jointly holomorphic. It satisfies

$$\mathcal{F}_{a,b,c}(s) = \frac{R_{a,b}(s) - 1}{s} - \left(\zeta(1+s, c) - \frac{1}{s} \right). \quad (6.5)$$

Both quotients on the right have their removable values at $s = 0$. Consequently, every finite mixed derivative in a, b, c, s may be taken term by term in (6.4).

Proof. The gamma-ratio expansion gives, uniformly on any compact subset of the parameter domain,

$$W_n(a, b; s) = n^{-1-s} (1 + O(n^{-1})), \quad (n+c)^{-1-s} = n^{-1-s} (1 + O(n^{-1})).$$

For precision concerning uniformity, apply the logarithmic Stirling expansion on a fixed sector containing all sufficiently large $n+a$, $n+b$, $n+a+b+s$, and $n+1$. On a compact parameter set the shifts are bounded, and the remainder after its first correction is uniform. Exponentiating gives the displayed estimates; this is the standard gamma-ratio expansion recorded in DLMF 5.11.13. If $\Re s \geq -1 + \eta$ on the compact set, with $\eta > 0$, the difference is $O(n^{-1-\eta})$. Thus the series converges normally and defines a jointly holomorphic function. Cauchy's formula on slightly larger compact polydiscs justifies all finite parameter derivatives.

For $\Re s > 0$, both series can be summed separately. Gauss summation, DLMF 15.4.20, gives

$$\sum_{n=0}^{\infty} W_n(a, b; s) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b+s)} {}_2F_1(a, b; a+b+s; 1) = \frac{\Gamma(a)\Gamma(b)\Gamma(s)}{\Gamma(a+s)\Gamma(b+s)}.$$

Subtract $\zeta(1+s, c)$ and use $\Gamma(1+s) = s\Gamma(s)$. Since $R_{a,b}(0) = 1$, both sides extend holomorphically through zero. The identity theorem in s proves (6.5) throughout $\Re s > -1$ for every permitted a, b, c . Joint holomorphy has already been established by normal convergence. $\square$

This is an ordinary convergent difference series. Splitting the two summands into separately divergent series at $s = 0$ would lose the specified normalization. The arbitrary comparison shift c is useful: the gamma-ratio coefficient does not change, while its logarithmic comparison term selects the generalized constant $\gamma_k(c)$.

6.2 Every spectral derivative in Bell-polynomial form

Let Y_k denote the complete exponential Bell polynomial, defined by

$$\exp\left(\sum_{j \geq 1} x_j \frac{t^j}{j!}\right) = \sum_{k \geq 0} Y_k(x_1, \dots, x_k) \frac{t^k}{k!}. \quad (6.6)$$

In particular $Y_0 = 1$, $Y_1 = x_1$, $Y_2 = x_1^2 + x_2$, and $Y_3 = x_1^3 + 3x_1x_2 + x_3$. Write

$$\kappa_j(a, b) = \psi^{(j-1)}(1) - \psi^{(j-1)}(a) - \psi^{(j-1)}(b), \quad j \geq 1. \quad (6.7)$$

Theorem 6.2 (All Stieltjes indices). *For every integer $k \geq 0$,*

$$\begin{aligned} \sum_{n=0}^{\infty} \left[W_n(a, b; 0) Y_k(-\psi(n+a+b), \dots, -\psi^{(k-1)}(n+a+b)) - \frac{(-\log(n+c))^k}{n+c} \right] \\ = \frac{Y_{k+1}(\kappa_1, \dots, \kappa_{k+1})}{k+1} - (-1)^k \gamma_k(c). \end{aligned} \quad (6.8)$$

For $k = 0$, the Bell factor in the summand is 1 and has no arguments. The series is absolutely convergent, locally uniformly in the shift parameters. Its summand is $O((1 + \log n)^k/n^2)$ on real compact parameter sets.

Proof. The k th s derivative of W_n at zero is the first term in brackets, because the j th logarithmic derivative of its sole s -dependent gamma factor is $-\psi^{(j-1)}(n+a+b)$. On the other hand,

$$R_{a,b}^{(k+1)}(0) = Y_{k+1}(\kappa_1, \dots, \kappa_{k+1}), \quad \left. \frac{d^k}{ds^k} \frac{R_{a,b}(s) - 1}{s} \right|_{s=0} = \frac{R_{a,b}^{(k+1)}(0)}{k+1}.$$

Differentiate (6.5) using (6.1). Normal convergence follows from Theorem 6.1. The explicit logarithmic summand estimate follows by differentiating the locally uniform gamma-ratio expansion, or by using the corresponding polygamma expansions. $\square$

The first three identities are

$$\sum_{n \geq 0} \left(W_n - \frac{1}{n+c} \right) = \kappa_1 - \gamma_0(c), \quad (6.9)$$

$$\sum_{n \geq 0} \left(W_n \psi(n+a+b) - \frac{\log(n+c)}{n+c} \right) = -\frac{\kappa_1^2 + \kappa_2}{2} - \gamma_1(c), \quad (6.10)$$

$$\sum_{n \geq 0} \left(W_n (\psi(n+a+b)^2 - \psi'(n+a+b)) - \frac{\log^2(n+c)}{n+c} \right) = \frac{\kappa_1^3 + 3\kappa_1\kappa_2 + \kappa_3}{3} - \gamma_2(c), \quad (6.11)$$

where $W_n = W_n(a, b; 0)$.

6.3 A complete mixed-parameter coefficient rule

The finite-dimensional Bell expression has a direct multivariate version. Put $q_n = n + a + b$ and introduce formal increments u, v, s . Define

$$\mathcal{K}_n(u, v, s) = \sum_{j \geq 1} \frac{1}{j!} \left[\psi^{(j-1)}(n+a)u^j + \psi^{(j-1)}(n+b)v^j - \psi^{(j-1)}(q_n)(u+v+s)^j \right], \quad (6.12)$$

$$\mathcal{R}(u, v, s) = \frac{\Gamma(1+s)\Gamma(a+u)\Gamma(b+v)}{\Gamma(a+u+s)\Gamma(b+v+s)}. \quad (6.13)$$

Its logarithm is the explicit formal series

$$\log \mathcal{R}(u, v, s) = \sum_{j \geq 1} \frac{1}{j!} [\psi^{(j-1)}(1)s^j + \psi^{(j-1)}(a)\{u^j - (u+s)^j\} + \psi^{(j-1)}(b)\{v^j - (v+s)^j\}]. \quad (6.14)$$

For fixed indices, only finitely many terms of these exponentials are needed.

Theorem 6.3 (All mixed jets). *For all integers $p, q, k \geq 0$,*

$$\begin{aligned} \sum_{n \geq 0} \left[W_n(a, b; 0) [u^p v^q s^k] e^{\mathcal{K}_n(u, v, s)} - \delta_{p0} \delta_{q0} \frac{(-\log(n+c))^k}{k!(n+c)} \right] \\ = [u^p v^q s^{k+1}] \mathcal{R}(u, v, s) - \delta_{p0} \delta_{q0} \frac{(-1)^k \gamma_k(c)}{k!}. \end{aligned} \quad (6.15)$$

All series are ordinary absolutely convergent series. Multiplying by $p!q!k!$ converts this coefficient formula into the mixed derivative formula. When $p+q > 0$, no logarithmic subtraction and no Stieltjes constant remain.

Proof. Taylor expansion of the three nonconstant gamma factors in $W_n(a+u, b+v; s)$ gives $W_n(a+u, b+v; s) = W_n(a, b; 0) e^{\mathcal{K}_n(u, v, s)}$. Apply Theorem 6.1 at $(a+u, b+v, c)$ and compare the indicated coefficient. The identity $\mathcal{R}(u, v, 0) = 1$ explains the shift from s^k to s^{k+1} without any division ambiguity. Normal convergence justifies the coefficient extraction under the sum. $\square$

One useful consequence is a cancellation at every positive mixed tangential order:

$$\sum_{n \geq 0} \partial_a^p \partial_b^q W_n(a, b; 0) = 0, \quad p, q \geq 1. \quad (6.16)$$

Indeed, the s^1 coefficient of $\mathcal{R}$ is $\psi(1) - \psi(a+u) - \psi(b+v)$, which has no mixed u, v terms. The pure derivatives satisfy

$$\sum_{n \geq 0} \partial_a^p W_n(a, b; 0) = -\psi^{(p)}(a), \quad p \geq 1, \quad (6.17)$$

and the analogous identity holds for b .

For an explicit quadratic identity, set $A_n = \psi(n+a) - \psi(q_n)$ and $B_n = \psi(n+b) - \psi(q_n)$. Then

$$\sum_{n \geq 0} W_n(a, b; 0) (A_n B_n - \psi'(q_n)) = 0. \quad (6.18)$$

The first spectral lift is also closed:

$$\sum_{n \geq 0} W_n(a, b; 0) [-\psi(q_n)\{A_n B_n - \psi'(q_n)\} - \psi'(q_n)(A_n + B_n) - \psi''(q_n)] = \psi'(a)\psi'(b). \quad (6.19)$$

This follows either from (6.15) at $(p, q, k) = (1, 1, 1)$ or by differentiating (6.10) with its original negative spectral sign retained. At $a = b = 1$, (6.18) is the classical $\zeta(2, 1) = \zeta(3)$ identity written as a polygamma series. We therefore do not present that specialization as a new evaluation.

6.4 Ordinary and central-binomial Stieltjes series

Let $H_n^{(r)} = \sum_{j=1}^n j^{-r}$, with $H_0^{(r)} = 0$, and write $H_n = H_n^{(1)}$. At $a = b = c = 1$ the weight is $1/(n+1)$ and $(\kappa_1, \kappa_2, \kappa_3) = (\gamma, -\zeta(2), 2\zeta(3))$. Equations (6.10)–(6.11) become

$$\sum_{m=1}^{\infty} \frac{H_m - \gamma - \log m}{m} = \frac{\zeta(2) - \gamma^2}{2} - \gamma_1, \quad (6.20)$$

$$\sum_{m=1}^{\infty} \frac{(H_m - \gamma)^2 + H_m^{(2)} - \zeta(2) - \log^2 m}{m} = \frac{\gamma^3 - 3\gamma\zeta(2) + 2\zeta(3)}{3} - \gamma_2. \quad (6.21)$$

The all-order form follows from (6.8); for $j \geq 2$ its local Bell arguments are $-\psi^{(j-1)}(m+1) = (-1)^{j+1}(j-1)!\{\zeta(j) - H_m^{(j)}\}$.

A less elementary weight is obtained at $a = b = \frac{1}{2}$, $c = 1$. Put

$$d_n = \frac{\binom{2n}{n}^2}{16^n}, \quad L = \log 2, \quad K = \gamma + 4L.$$

The duplication formula gives $W_n = \pi d_n$, while

$$\kappa_1 = K, \quad \kappa_j = (3 - 2^{j+1})(-1)^j(j-1)!\zeta(j) \quad (j \geq 2). \quad (6.22)$$

Thus

$$\sum_{n=0}^{\infty} \left(\pi d_n - \frac{1}{n+1} \right) = 4 \log 2, \quad (6.23)$$

$$\sum_{n=0}^{\infty} \left(\pi d_n (H_n - \gamma) - \frac{\log(n+1)}{n+1} \right) = \frac{5}{2}\zeta(2) - \frac{K^2}{2} - \gamma_1, \quad (6.24)$$

$$\sum_{n=0}^{\infty} \left(\pi d_n \{ (H_n - \gamma)^2 + H_n^{(2)} - \zeta(2) \} - \frac{\log^2(n+1)}{n+1} \right) = \frac{K^3}{3} - 5K\zeta(2) + \frac{26}{3}\zeta(3) - \gamma_2. \quad (6.25)$$

For every k , the same squared central-binomial weight gives γ_k through (6.8) with (6.22); no empirical relation search is involved.

The tangential identities yield convergent harmonic sums without a Stieltjes subtraction. Since $\psi(n + \frac{1}{2}) - \psi(n+1) = 2(H_{2n} - H_n - L)$, they imply

$$\sum_{n=0}^{\infty} d_n (H_{2n} - H_n - L) = -\frac{\pi}{4}, \quad (6.26)$$

$$\sum_{n=0}^{\infty} d_n \left[4(H_{2n} - H_n - L)^2 + H_n^{(2)} - \zeta(2) \right] = 0. \quad (6.27)$$

The brackets must remain grouped: some of their separately summed pieces diverge. The summands displayed above are $O(n^{-2})$.

6.5 Polylogarithmic boundary values

For $|z| < 1$, define

$$\begin{aligned} \mathcal{F}_{a,b,c}(s; z) &= \sum_{n \geq 0} \left(W_n(a, b; s) - (n+c)^{-1-s} \right) z^n \\ &= \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b+s)} {}_2F_1(a, b; a+b+s; z) - \Phi(z, 1+s, c). \end{aligned} \quad (6.28)$$

At lower hypergeometric parameters that are nonpositive integers, the first expression specifies the regularized meaning of the normalized hypergeometric factor. Uniform absolute convergence on $|z| \leq 1$, locally in $\Re s > -1$, proves the exact boundary identity

$$\lim_{z \rightarrow 1, |z| \leq 1} \partial_a^p \partial_b^q \partial_s^k \mathcal{F}_{a,b,c}(s; z) = \partial_a^p \partial_b^q \partial_s^k \mathcal{F}_{a,b,c}(s). \quad (6.29)$$

The limit is not restricted to a radial approach. In particular, for $c = 1$ the Lerch factor is

$$\Phi(z, 1+s, 1) = \frac{\text{Li}_{1+s}(z)}{z}, \quad (6.30)$$

so (6.8) is also a family of exact boundary constants for polylogarithm order derivatives after an explicit hypergeometric subtraction.

For rational $c = r/q$, $1 \leq r \leq q$, let $\omega = e^{2\pi i/q}$ and, for example, take $0 < z < 1$. The root-of-unity filter gives

$$\Phi(z, 1 + s, r/q) = q^s z^{-r/q} \sum_{j=0}^{q-1} \omega^{-jr} \operatorname{Li}_{1+s}(\omega^j z^{1/q}). \quad (6.31)$$

Indeed the sum selects positive exponents congruent to r modulo q ; after writing those exponents as $qn + r$, the prefactor becomes exactly $(n + r/q)^{-1-s}$. Finite translation reduces every positive rational c to this range. All s derivatives bring only finite binomial combinations with powers of $\log q$, in addition to polylogarithm order derivatives.

At the central-binomial point $s = 0$, the first term in (6.28) is a classical complete elliptic integral $\pi {}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; z)$. It is twice the complete elliptic integral with parameter z . Consequently (6.23) is its logarithmic boundary constant, whereas (6.24) and (6.25) describe the next spectral layers of the same cancellation.

7 Successive subtractions and every negative balance

The first subtraction in Theorem 6.1 has an exact extension through every negative integer balance. Its coefficients are finite polynomials, and residue cancellation gives an identity among those polynomials that is independent of the comparison shift.

Let $B_j(x)$ be the Bernoulli polynomials, defined by

$$\frac{te^{xt}}{e^t - 1} = \sum_{j \geq 0} B_j(x) \frac{t^j}{j!}.$$

For $j \geq 1$, set

$$D_j(a, b, s; c) = \frac{(-1)^{j+1}}{j(j+1)} [B_{j+1}(a-c) + B_{j+1}(b-c) - B_{j+1}(a+b+s-c) - B_{j+1}(1-c)]. \quad (7.1)$$

Define polynomials $A_j = A_j(a, b, s; c)$ by the formal identity

$$\sum_{j \geq 0} A_j t^j = \exp \left(\sum_{j \geq 1} D_j t^j \right). \quad (7.2)$$

Equivalently,

$$A_0 = 1, \quad A_m = \frac{1}{m} \sum_{j=1}^m j D_j A_{m-j} \quad (m \geq 1). \quad (7.3)$$

This definition is purely finite for any fixed m . The series in (7.2) is used formally; it need not converge as an infinite series in t .

Theorem 7.1 (Exact continuation by successive coefficient subtraction). *Fix an integer $M \geq 0$. On the domain*

$$\Re a > 0, \quad \Re b > 0, \quad \Re c > 0, \quad \Re s > -M - 1,$$

the series

$$\mathcal{F}_{a,b,c}^{[M]}(s) = \sum_{n=0}^{\infty} \left[W_n(a, b; s) - \sum_{j=0}^M A_j(a, b, s; c) (n+c)^{-1-s-j} \right] \quad (7.4)$$

converges normally on compact subsets and is jointly holomorphic. It equals the holomorphic continuation of

$$\frac{\Gamma(a)\Gamma(b)\Gamma(s)}{\Gamma(a+s)\Gamma(b+s)} - \sum_{j=0}^M A_j(a, b, s; c) \zeta(1+s+j, c). \quad (7.5)$$

Every apparent pole at $s = 0, -1, \dots, -M$ is removable in this complete expression. For every integer $r \geq 0$, the finite polynomial identity

$$A_r(a, b, -r; c) = \frac{(-1)^r \Gamma(a) \Gamma(b)}{r! \Gamma(a-r) \Gamma(b-r)} = \frac{(-1)^r}{r!} (1-a)_r (1-b)_r \quad (7.6)$$

holds. In particular its left side is independent of c .

Proof. Write $x = n + c$. The locally uniform logarithmic gamma expansion in a fixed sector around the positive real axis gives

$$\log W_n(a, b; s) = -(1+s) \log x + \sum_{j=1}^M \frac{D_j(a, b, s; c)}{x^j} + O(x^{-M-1}). \quad (7.7)$$

For all sufficiently large n the gamma factors have no zeros or poles on the compact parameter set, so a logarithm may be chosen consistently there. The coefficient formula follows directly by substituting the four shifts $a - c, b - c, a + b + s - c, 1 - c$ into the logarithmic gamma expansion in Bernoulli polynomials. Its leading powers combine to $-(1+s) \log x$, while the constant terms cancel. The remainder is uniform on compact parameter sets: all four shifts are bounded, and all four large gamma arguments lie in a common closed sector that avoids the negative real axis.

Exponentiating (7.7) to finite order and using (7.2) yields

$$W_n(a, b; s) = x^{-1-s} \left(\sum_{j=0}^M A_j(a, b, s; c) x^{-j} + O(x^{-M-1}) \right).$$

Thus the bracket in (7.4) is $O(n^{-M-2-\Re s})$. On a compact subset of the stated domain it is $O(n^{-1-\eta})$ for some $\eta > 0$. This proves normal convergence and joint holomorphy. The finitely many small n cause no difficulty: W_n is defined using the entire reciprocal gamma factor in (6.2), including at its zeros.

For $\Re s > 0$, every series can be summed separately and Gauss summation gives (7.5). Its meromorphic continuation in the larger half-plane has possible poles only at $s = 0, -1, \dots, -M$. The identity theorem, first away from those points and then across them, identifies it with the holomorphic left side. All its residues there must vanish. At $s = -r$, only the term indexed by $j = r$ in the zeta sum can have a pole, and its residue is $A_r(a, b, -r; c)$. The gamma quotient has residue

$$\frac{(-1)^r \Gamma(a) \Gamma(b)}{r! \Gamma(a-r) \Gamma(b-r)}.$$

This expression is interpreted through reciprocal gamma if either denominator argument is a nonpositive integer; its value is then zero, and the corresponding gamma pole is canceled. Equating the two residues proves the first equality in (7.6). The gamma recurrence proves the polynomial expression in the second equality. Since M can be any integer at least r , the result holds for every r . $\square$

The quotient in (7.6) is holomorphic on the stated a, b domain even when $a - r$ or $b - r$ is a nonpositive integer. The Pochhammer expression makes this particularly transparent. One must not replace such a reciprocal gamma factor by an undefined division by an infinite gamma value, nor differentiate a logarithm through one of its zeros.

7.1 An exact jet formula at a negative integer

Fix $0 \leq r \leq M$. Suppress a, b, c from the notation for A_j and write $A_j^{(k)}$ for its k th derivative in s . Define the analytic germ

$$\mathcal{R}_r(\varepsilon) = \frac{\Gamma(1+\varepsilon) \Gamma(a) \Gamma(b)}{\prod_{h=1}^r (\varepsilon - h)} \frac{1}{\Gamma(a-r+\varepsilon) \Gamma(b-r+\varepsilon)}. \quad (7.8)$$

The empty product is 1. This germ is analytic for sufficiently small ε , also when either reciprocal gamma factor vanishes at zero. It equals ε times the meromorphic gamma quotient in (7.5) evaluated at $s = -r + \varepsilon$.

Corollary 7.2 (Every resonant spectral jet). *For each integer $k \geq 0$,*

$$\begin{aligned} \left(\mathcal{F}_{a,b,c}^{[M]}\right)^{(k)}(-r) &= \frac{\mathcal{R}_r^{(k+1)}(0) - A_r^{(k+1)}(-r)}{k+1} \\ &\quad - \sum_{\ell=0}^k \binom{k}{\ell} A_r^{(k-\ell)}(-r) (-1)^\ell \gamma_\ell(c) \\ &\quad - \sum_{\substack{0 \leq j \leq M \\ j \neq r}} \sum_{\ell=0}^k \binom{k}{\ell} A_j^{(k-\ell)}(-r) \zeta^{(\ell)}(1-r+j, c). \end{aligned} \quad (7.9)$$

The derivatives on the left are given by termwise derivatives of the convergent brackets in (7.4). This formula remains valid at all admissible a, b, c , including the parameter values at which a gamma factor in an unregularized logarithmic expression would have a pole.

Proof. Put $Z(\varepsilon, c) = \zeta(1 + \varepsilon, c) - 1/\varepsilon$. Equations (7.5) and (7.8) give the exact germ

$$\begin{aligned} \mathcal{F}^{[M]}(-r + \varepsilon) &= \frac{\mathcal{R}_r(\varepsilon) - A_r(-r + \varepsilon)}{\varepsilon} - A_r(-r + \varepsilon) Z(\varepsilon, c) \\ &\quad - \sum_{j \neq r} A_j(-r + \varepsilon) \zeta(1 - r + j + \varepsilon, c). \end{aligned}$$

The numerator of the first term vanishes at zero by (7.6). Taking k derivatives and using (6.1) gives the formula. Normal convergence on slightly larger compact polydiscs justifies its series version. $\square$

If neither $a - r$ nor $b - r$ is a nonpositive integer, put $Q_r = A_r(a, b, -r; c)$. The finite value $k = 0$ can then be written

$$\begin{aligned} \mathcal{F}^{[M]}(-r) &= Q_r(\psi(r+1) - \psi(a-r) - \psi(b-r)) - A'_r(-r) - Q_r \gamma_0(c) \\ &\quad - \sum_{j \neq r} A_j(-r) \zeta(1 - r + j, c). \end{aligned} \quad (7.10)$$

At the excluded logarithmic parameter values the analytic germ (7.8), rather than an informal $0 \cdot \infty$ convention, specifies the value. Higher derivatives of this germ can also be written as Bell polynomials. When $Q_r \neq 0$, its logarithmic arguments are

$$\psi^{(j-1)}(1) + (j-1)! H_r^{(j)} - \psi^{(j-1)}(a-r) - \psi^{(j-1)}(b-r), \quad j \geq 1.$$

7.2 An exact central-binomial identity at balance minus one

Take $a = b = 1/2$, $c = 1$, $M = r = 1$. Formula (7.1) gives

$$A_1(s) = \frac{3}{4} + \frac{s}{2} - \frac{s^2}{2}, \quad A_1(-1) = -\frac{1}{4}, \quad A'_1(-1) = \frac{3}{2}.$$

Moreover,

$$W_n(1/2, 1/2; -1) = \pi n \frac{\binom{2n}{n}^2}{16^n}.$$

At $n = 0$ this equality has its literal value 0: the original gamma ratio contains $1/\Gamma(0) = 0$, interpreted through the entire reciprocal gamma function. The rest of the $n = 0$ bracket remains $-1 + 1/4 = -3/4$ and is included in the sum.

Using $\psi(2) = 1 - \gamma$, $\psi(-1/2) = 2 - \gamma - 2\log 2$, $\zeta(0) = -1/2$, and $\gamma_0(1) = \gamma$ in (7.10) yields the exact identity

$$\sum_{n=0}^{\infty} \left[\pi n \frac{\binom{2n}{n}^2}{16^n} - 1 + \frac{1}{4(n+1)} \right] = -\log 2 - \frac{1}{4}. \quad (7.11)$$

The bracket is $O(n^{-2})$. Its three displayed pieces must be summed together; the individual series do not converge.

The numerical verification evaluates this bracket directly, then uses the independent gamma-ratio asymptotic tail

$$\sum_{j=2}^K A_j(1/2, 1/2, -1; 1) \zeta(j, N+1).$$

It checks stability under two choices of N, K . The same script verifies (7.6) symbolically for $r = 0, 1, 2, 3, 4$ with all of a, b, c indeterminate. These computational checks support, and are separate from, the all-order proof above.

8 Correcting and proving a historical family of harmonic conjectures

The paper of Choi [16] uses the standard definition $H_n = \sum_{j=1}^n 1/j$. Its printed page 6 contains Conjecture 4, Corollary 5, and Conjecture 6. The formulas can be checked and extended uniformly by the following generating identity. This discussion concerns those displayed statements, and does not assert that they remain open or that no correction has appeared elsewhere.

Theorem 8.1 (A two-parameter harmonic generating identity). *For u, v in a neighborhood of zero,*

$$\sum_{n=0}^{\infty} \frac{(1+u)_n(1+v)_n}{(n!)^2(n+1)(n+2)} = \frac{\Gamma(1-u-v)}{\Gamma(2-u)\Gamma(2-v)}. \quad (8.1)$$

Define

$$E_r(n) = Y_r(H_n, -H_n^{(2)}, 2H_n^{(3)}, \dots, (-1)^{r-1}(r-1)!H_n^{(r)}), \quad E_0(n) = 1. \quad (8.2)$$

Then

$$\sum_{n=1}^{\infty} \frac{E_r(n)}{(n+1)(n+2)} = r!, \quad r \geq 1, \quad (8.3)$$

$$\sum_{n=1}^{\infty} \frac{H_n E_r(n)}{(n+1)(n+2)} = r! \left(1 + \sum_{j=2}^{r+1} \zeta(j) \right), \quad r \geq 0. \quad (8.4)$$

Every mixed sum with $E_r(n)E_t(n)$ is likewise given by the finite Taylor coefficient $r!t![u^r v^t]$ of the gamma quotient in (8.1), except that the $r = t = 0$ formula must retain the $n = 0$ term or subtract its value $1/2$.

Proof. Since $(3)_n = n!(n+1)(n+2)/2$, the series in (8.1) is $\frac{1}{2} {}_2F_1(1+u, 1+v; 3; 1)$. Gauss summation proves the stated gamma quotient. Uniform convergence and differentiation follow, for instance, from $O(n^{-2+\Re(u+v)})$ on a small closed polydisc around zero. The coefficient identity

$$\frac{(1+u)_n}{n!} = \prod_{j=1}^n \left(1 + \frac{u}{j} \right) = \sum_{r \geq 0} \frac{E_r(n)}{r!} u^r$$

follows from (6.6); it also shows that $E_r(n)$ is $r!$ times an elementary symmetric polynomial in $1, 1/2, \dots, 1/n$.

Set $v = 0$ in (8.1). Its right side becomes $1/(1 - u)$, which proves (8.3). For the second identity, differentiate in v at zero to obtain

$$\partial_v \frac{\Gamma(1 - u - v)}{\Gamma(2 - u)\Gamma(2 - v)} \Big|_{v=0} = \frac{\psi(2) - \psi(1 - u)}{1 - u} = \frac{1 + \sum_{j \geq 1} \zeta(j + 1)u^j}{1 - u}.$$

Taking the coefficient of u^r proves (8.4). The zero-th summand has zero derivative in every case used in those two identities, which explains their lower limit $n = 1$. $\square$

In the notation of Theorem 5.1, $E_r(n) = r!E_{n,r}$. The pure identity also follows independently from the resolvent difference at $a = 1, b = 2$, since $F(1, y) = 1$ and $F(2, y) = (1 - y)^{-1}$. The product with H_n requires the two-parameter coefficient in the theorem.

The theorem gives the intended fifth- and sixth-order cases of Conjecture 4 through (8.3). It also gives corrected versions of all six displayed formulas (41)–(46) through (8.4). The precise corrections are:

1. The right sides in (41)–(46) must be divided by two; for (46), the numerator correction below is also required. With those corrections, the multiplier in the row indexed by r is $r!$, for $r = 1, \dots, 6$, instead of the printed $2r!$.
2. In the second identity of (40), the final harmonic term in the sixth-order Bell polynomial must be $-120H_n^{(6)}$; the printed version has $-120H_n^{(5)}$.
3. In (46), the term belonging to the partition $2 + 2 + 1 + 1$ must be $45H_n^3(H_n^{(2)})^2$ after the outer factor H_n is included. The printed formula has $45H_n^3(H_n)^2$ at that position.

These readings were checked against the publisher's PDF, not inferred from its mathematical-text extraction. The useful correction is the unambiguous Bell-polynomial statement, which also fixes all higher orders without enumerating partitions individually.

The simplest factor-two error admits a short proof independent of hypergeometric differentiation. By summation by parts and Tonelli's theorem for the nonnegative terms,

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{H_n^2}{(n+1)(n+2)} &= \sum_{k=1}^{\infty} \frac{H_k^2 - H_{k-1}^2}{k+1} \\ &= 2 \sum_{k=1}^{\infty} \frac{H_{k-1}}{k(k+1)} + \sum_{k=1}^{\infty} \frac{1}{k^2(k+1)} \\ &= 2 \sum_{j=1}^{\infty} \frac{1}{j(j+1)} + \zeta(2) - 1 = 1 + \zeta(2). \end{aligned}$$

This disproves the printed value $2(1 + \zeta(2))$ in (41) with its stated definitions. It also isolates the origin of the normalization: the hypergeometric function is twice the left side of (8.1), rather than equal to it.

9 Source audit, proof checks, and reproducibility

9.1 The five incoming reports

The source audit distinguishes the following archives at the pinned commit. The short names in this section are only reading aids; the machine-readable record preserves every full filename and internal path.

RC `ProveIt_Resonance_Correlations_2026-10-10.zip`. Existing Gamma and Hurwitz-jet correlations supply background. The present triple formula answers the coordinate part of item G3 in `sections/07_questions.tex`; minimal character spaces and arithmetic independence remain separate.

- RJ** ProveIt_Shifted_Hurwitz_Jets_2026-10-10.zip. Its bilinear kernel and unit-scale contact terms are retained. The questions on three shifted factors and on dilation, traces, and subtraction scales in `sections.tex` are answered here under the stated separation and positive-integer hypotheses.
- SC** ProveIt_Stieltjes_Convolution_2026-10-10.zip. Its periodic convolution algebra and one-factor derivative anomalies are established input. The affine-frequency future-work sentence requires the qualification described below.
- CL** ProveIt_Stieltjes_Correlation_Closure_2026-10-10.zip. Its all-index bilinear linear closure and missing-order rule are rederived compactly in Theorem 3.3. The trilinear and unequal-integer-dilation questions in its final section are answered for separated singularities.
- CC** Stieltjes_Convolution_Calculus_2026-10-10.zip. Its derivative and primitive ladders overlap the other reports. The first further-research item, concerning a triple generating function and subtraction rule, is answered here; the period and coordinate question is answered for affine integer rescaling.

The exact anchors and Git blob identifiers appear in `integration/source_audit.json`. Overlap is recorded as established background, not counted again as a present result.

9.2 Proposed source corrections

The only definite correction proposed to the reviewed repository text is to SC’s future-work discussion, `article.tex`, lines 1118–1121. Two nonzero integer affine frequencies leave one free integer in $pn + qm = 0$. Thus a different integer frequency alone does not force a higher-dimensional summation index. The replacement text in `integration/proposed_corrections.tex` states the precise bilinear theorem and reserves the higher-dimensional issue for three or more independent factors. This correction does not invalidate SC’s convolution identities.

The external corrections to Choi’s printed p. 6 are mathematically different: the displayed identities (41)–(46), with the stated definitions, have twice the correct right side, and two harmonic orders are misprinted. Theorem 8.1 gives corrected formulas and a proof for every order. The correction record includes the DOI, page, and equation numbers. The publisher PDF was visually checked. Neither the absence of a located erratum nor numerical agreement is used to infer historical priority.

The independent proof audit of this article checked the Fourier phase convention, the six disjoint frequency cones, the pole residues, the Gamma and harmonic factorials, and the mixed parameter domains. It also sharpened the uniform local estimate used for primitive correlations and the Cauchy argument at simultaneous spectral poles. These improvements are incorporated into the proofs as delivered.

9.3 The surviving named Gaussian conjectures

The manuscript’s names S_6 and S_8 refer to specific alternating harmonic sums, and should not be confused with the uncolored sums of our Gamma-ratio theorem. With the largest summation index first in the multiple-polylogarithm convention, their definitions are

$$S_p = \sum_{n \geq 0} \frac{(-1)^n H_n}{(2n+1)^p}, \quad g_{a,b} = \operatorname{Im} \operatorname{Li}_{a,b}(i, 1) = \sum_{n \geq 0} \frac{(-1)^n H_{2n}^{(b)}}{(2n+1)^a},$$

where $H_m^{(b)} = \sum_{j=1}^m j^{-b}$ and $H_m = H_m^{(1)}$. Write $G = \beta(2)$ for Catalan’s constant. The existing mixed-endpoint identity gives

$$S_p = g_{p,1} + \operatorname{Im} \operatorname{Li}_{p,1}(i, -1).$$

In particular, S_p is not just $g_{p,1}$.

The frozen weight-seven conjecture currently reads

$$S_6 \stackrel{?}{=} \frac{722}{527}g_{6,1} + \frac{40}{527}g_{4,3} - \frac{128}{155}g_{2,5} + \frac{15191}{28569600}\pi^7 \\ - \frac{3}{124}G\zeta(5) - \frac{2373}{10540}\beta(4)\zeta(3) - 2\beta(6)\log 2. \quad (9.1)$$

The source is `chapters/04-cyclotomic-quotients.tex`, lines 338–349. Its label is `cycloquot:conj:S6`. The alternative formula involving $g_{5,2}$ is already proved equivalent to this one by two convergent shuffle identities in the same file, lines 295–336. That coordinate change does not prove either conjectured vanishing. Moreover, the source’s separating functional is nonzero on the frozen relation, so a proof cannot use only the restricted single-product presentation considered there.

The current weight-nine conjecture is

$$S_8 \stackrel{?}{=} \frac{52334}{37973}g_{8,1} + \frac{2824}{37973}g_{6,3} - \frac{6832}{189865}g_{4,5} - \frac{139760}{265811}g_{2,7} \\ + \frac{998183\pi^9}{17283022848} - \frac{78615}{19442176}G\zeta(7) - \frac{201021}{4252976}\beta(4)\zeta(5) \\ - \frac{290505}{1063244}\beta(6)\zeta(3) - 2\beta(8)\log 2. \quad (9.2)$$

Its exact source is `chapters/04-S8-candidate.tex`, lines 7–17, label `s8new:conj:S8`. The same file, lines 40–50, records exact rational proximity bounds below 10^{-355} for the normalized residuals of both current conjectures. Both enclosures contain zero. The equations remain conjectural in the pinned manuscript and in the present report.

This S_8 candidate must be distinguished from the earlier frozen vector in the basket containing $g_{7,2}$ and $g_{5,4}$. That different vector has a strictly negative certified residual between $-1.3274179020580459 \times 10^{-86}$ and $-1.3274179020580457 \times 10^{-86}$; see `chapters/10-discovery.tex`, lines 259–301, label `research:prop:S8-rejected`. The rejection concerns precisely that vector and gives no arithmetic independence result.

Our uncolored harmonic theorem does not acquire the factor $(-1)^n$ merely by setting its Hurwitz parameter to $1/2$. Its direct application therefore cannot prove either displayed Gaussian identity. The trilinear theorem does provide root-of-unity colored Tornheim coordinates at rational shifts, with multiple-polylogarithm reductions at positive integer inner orders. A coordinate representation alone does not annihilate either frozen residual, and the partial-fraction identities for discrete inner orders cannot be differentiated as though they held with a continuously varying summation range. The remaining concrete problem is to find and prove the missing mixed-endpoint functional relation at weights seven and nine, and then reduce it exactly to the specified frozen basket.

9.4 Independent numerical diagnostics

The programs evaluate two mathematically different sides whenever practical. They are numerical diagnostics, not interval certificates; the proofs in the preceding sections establish the identities. Working precision is larger than the number of digits reported.

Component	Independent comparison	Largest observed residual
Integer dilations	Local digamma subtraction against the scale formula; four frequency pairs at 60 digits	7.4×10^{-58}
Ordinary log-Gamma	Direct split quadrature against polylogarithmic order finite differences, step 10^{-7}	1.9×10^{-31}
Triple products	Local three-digamma subtraction against a separate Mellin and polylogarithm-order evaluation, at 42 working digits	8.7×10^{-42}
Shifted harmonic sums	37 Mellin and local-subtraction comparisons against explicit polygamma formulas, at 70 digits	1.82×10^{-56}
Gauss mixed jets	48 direct sums with Bernoulli-polynomial tails against mixed Gamma coefficients, at 75 digits	5.14×10^{-58}
Negative balance	Direct central-binomial bracket with an independent tail, at 75 digits	1.97×10^{-58}

The reported maxima are rounded upward from the stored results. For the Gauss checks the change when the tail parameters are refined from $(N, K) = (70, 24)$ to $(110, 32)$ is below 6.00×10^{-42} . This refinement is useful evidence about numerical stability and is not itself a rigorous error estimate. The log-Gamma finite-difference comparison has deliberately lower accuracy than the digamma finite part; its error is dominated by the derivative approximation.

The dilation check computes the unit-frequency base integral by its own local subtraction. Consequently it checks the new frequency and scale formula without assuming the closed evaluation of J_{00} . The triple check additionally compares the six-cone formula at $\lambda = (1, 1, 1)$ and $(2, 1, 3)$, with rational shifts $0, 1/5, 7/10$, against exact piecewise Bernoulli-polynomial integration. The exact values are respectively $-1/1000$ and $-124409/7560000000$.

For the equal-thirds digamma example, the two numerical routes give

$$\text{FP} \int_0^1 \psi(x)\psi(\{x + \tfrac{1}{3}\})\psi(\{x + \tfrac{2}{3}\}) dx = 27.805592568749290182019060698117 \dots \quad (9.3)$$

This decimal is a numerical evaluation of the exact Tornheim formula (4.14), not a replacement for it.

9.5 Exact arithmetic and reproducible artifacts

The exact coefficient generator in `code/exact_coefficients.py` uses SymPy arithmetic to produce Gamma coefficients, rational-kernel reductions, local contact polynomials, bilinear Stieltjes coefficients, and the corrected two-parameter harmonic sums. Its inputs and outputs are exact; no integer-relation search is used. The formal Stieltjes symbols in a coefficient table are named atoms, with no assumed arithmetic independence. A separate exact six-cone check is explained in Appendix A.

The included scripts and JSON results make each numerical comparison reproducible. `README.md` gives the build and check commands, `requirements.txt` records the Python packages, and `MANIFEST.sha256` records checksums of the final files. `integration/INTEGRATION_NOTES.md` maps theorem labels to source questions and lists the necessary finite-part conventions. These files are a research supplement. No Lean, Isabelle, or other proof-assistant certification is claimed.

10 Further research questions and concrete next targets

The most useful continuations seek exact identities and reductions. The questions below distinguish a representation problem, a reduction problem in a prescribed constant space, and an arithmetic independence problem; solving the first does not automatically solve the other two.

10.1 The frozen Gaussian relations at weights seven and nine

The immediate named-conjecture target remains the exact vanishing of the current S6 and S8 residuals in (9.1)–(9.2). The mixed endpoint $\text{Li}_{p,1}(i, -1)$ is essential. A useful next result would be a functional identity involving both endpoints that is valid before specialization to a fixed weight. Its specialization could then be reduced by exact shuffle and stuffle relations to the frozen baskets. For S6, a candidate argument can be tested against the source’s separating functional: an argument confined to the already insufficient restricted presentation cannot suffice.

An exact residual derivation should record the initial analytic domain, the admissible endpoint continuation, every functional relation used, and the final rational coefficient vector. Recomputing a very small numerical residual does not address the missing functional identity. The colored kernels developed here can organize additional relations, but no identity annihilating those two residuals has been proved in this article.

10.2 Minimal coordinates for the first triple Stieltjes periods

Theorem 4.4 establishes a finite Tornheim-jet description at every index. The next question is whether, at rational shifts, symmetries and character decompositions reduce the number of independent formal jet coordinates. Start with the equal-thirds constant in (4.14) and the first indices $(1, 0, 0)$ and $(1, 1, 0)$. The integer-value identity at $C = 0$ determines $\mathcal{T}(A, B, 0) = \text{Li}_A(X) \text{Li}_B(Y)$, but it does not determine $\partial_C \mathcal{T}(A, B, 0)$; the latter contains a true logarithm of the coupled summation variable $m + n$.

A concrete deliverable would be a finite presentation of these low jet orders modulo the proved symmetries, conjugation, distribution, and functional equations, together with an exact reduction of the triple digamma expression in that presentation. A nonzero formal coordinate would indicate that more relations are needed. It would not prove arithmetic independence of the associated complex numbers.

10.3 Three unequal integer frequencies

With three affine frequencies, the constraint is $p_1 n_1 + p_2 n_2 + p_3 n_3 = 0$ and generally leaves a lattice of rank two. In each sign region, divisibility conditions arise when one frequency is eliminated. Finite character filters can express those conditions; the remaining denominators involve positive integer linear forms. The natural question is whether a finite family of weighted colored Tornheim kernels gives a canonical formula compatible with the local contact polynomial of Theorem 3.2.

The first target should keep all singular sets disjoint and prove the ordinary spectral identity in an absolutely convergent half-space. One must then show that the continued kernel gives the specified fixed-coordinate finite part. Merely solving the lattice equation is insufficient when the local scales differ. Four or more translated factors would introduce higher-dimensional analogues and a larger set of sign regions.

10.4 Collision identities and nonlinear coordinate changes

When two shifts coalesce, the disjoint-support product argument no longer applies. It is natural to compare three precisely defined operations: the separated-shift formula as the separation tends to zero, the finite part of the coincident product, and the regular coefficient of a joint spectral continuation. Exact difference formulas between these operations could identify the complete local counterterms without assuming they commute.

A related question replaces the affine coordinate pt by $pt + q_2 t^2 + q_3 t^3 + \dots$. For each fixed Stieltjes and derivative order, only finitely many Taylor coefficients of this coordinate can enter a distribution supported at the endpoint. Derive that finite law, recover $b_{n,k}(\log p)$ when the nonlinear

coefficients vanish, and check composition of coordinate changes exactly. Such a law would give a principled extension of the scale convention instead of adding unrelated constants to special cases.

10.5 Mixed harmonic Stieltjes data beyond the constant term

The moving-pole theorem determines $R_{r,0}(a)$ and the ordinary Stieltjes boundary $R_{0,k}(a)$. The genuinely mixed coefficients $R_{r,k}(a)$ with $r, k \geq 1$ remain a distinct question. A first target is $R_{1,1}(a)$ at $a = 1, 1/2$, and a general positive rational shift. Its convergent integral in (5.52) makes the normalization exact; what is missing is an additional reduction.

One can also introduce an independent shift in the inner harmonic denominators, or a color ξ^n in the outer sum. The beta/Gamma summation class then changes. Classify which choices still admit a Gamma-quotient generating theorem and which require hypergeometric, multiple-Lerch, or elliptic coordinates. This is especially relevant to avoiding the false inference that the uncolored half-shift formula already evaluates the alternating S6 and S8 sums.

10.6 Negative-balance identities and other summation theorems

The polynomial identity (7.6) is independent of the comparison shift c . A direct finite Bernoulli-polynomial proof, separate from the analytic residue proof, would explain this rigidity algebraically and could expose further parameter symmetries. Different values of M in (7.4) are related by explicit Hurwitz terms; determine a convenient minimal normalization for tables of the higher resonance jets.

The same program can be applied to balanced ${}_3F_2$ and other classical summation formulas. The issue is to identify the full asymptotic coefficient of each possible pole before taking parameter derivatives. Which summation theorems retain a finite Gamma-product answer after these subtractions? Which instead force multiple-zeta or higher hypergeometric periods? A useful result would classify one coherent family with a complete mixed-jet theorem, rather than enumerate a few unconnected low-order sums.

10.7 The full polylogarithmic boundary function

At rational c , equation (6.31) converts the comparison term to finitely many colored polylogarithms and their order derivatives. The other side, at $a = b = 1/2$, already involves a complete elliptic integral. Determine a natural function class for the full z -dependent mixed jets of (6.28), and formulate connection identities between $z = 0$ and $z = 1$ in that class.

The boundary constants proved here supply exact normalization data. They do not imply that the entire family reduces to ordinary multiple polylogarithms. A first concrete study could treat the first two order derivatives at the central-binomial point and compare their differential equations with the explicitly known Stieltjes boundary constants.

10.8 Exact symbolic generation and formal proof interfaces

The rational-kernel and contact-polynomial generators are small enough to serve as a first exact implementation target for ProveIt. The essential invariant for a rational kernel is the cancellation of the sum of its simple-pole residues; losing it would replace a convergent identity by separately divergent expressions. The essential invariant for a finite part is its named local coordinate and contact term.

A formal development could isolate reusable lemmas for locally normal parameter families, Mellin subtraction, meromorphic coefficient extraction, and products with disjoint singular supports. These lemmas would support every-order theorems without separate proofs for every table entry. Any computational search for additional constant relations should preserve an exact ordered basket and return a candidate status until an analytic derivation or an exact relation certificate is supplied.

A Exact coefficient checks and explicit correction polynomials

A.1 The leading triple pole

There is a simple exact check on the six-cone normalization. Put $z_j = e(a_j)$, with the three z_j distinct. Since $\text{Li}_0(z) = z/(1-z)$, equation (4.6) at zero gives

$$S(0, 0, 0; \mathbf{a}) = \sum_{k=1}^3 \frac{z_i z_j + z_k^2}{(z_k - z_i)(z_k - z_j)} = 2, \quad \{i, j, k\} = \{1, 2, 3\}. \quad (\text{A.1})$$

For an exact proof let $e_1 = z_1 + z_2 + z_3$ and $e_2 = z_1 z_2 + z_1 z_3 + z_2 z_3$. Then $z_i z_j = z_k^2 - e_1 z_k + e_2$. The Lagrange interpolation formulas for $1, t, t^2$ give respectively zero, zero, and one when their values at z_k are divided by $\prod_{j \neq k} (z_k - z_j)$ and summed. The displayed rational identity follows. Consequently the leading spectral pole is $2/(\lambda_1 \lambda_2 \lambda_3)$.

The same sign follows from the separated singular supports: $\int \prod_{j=1}^3 (\delta_{x_j} - 1) = 3 - 1 = 2$. This is a normalization check, not a substitute for the full coefficient-extraction proof.

A.2 First bilinear closure coefficients

Theorem 3.3 gives, with $b = 1 - c$,

$$J_{00}(c) = \gamma_1(c) + \gamma_1(b) - 2\zeta(2), \quad (\text{A.2})$$

$$J_{01}(c) = \gamma_2(c) + \frac{1}{2}\gamma_2(b) + \zeta(2)(\gamma_0(c) + \gamma_0(b)) - \zeta(3), \quad (\text{A.3})$$

$$J_{11}(c) = \frac{1}{2}(\gamma_3(c) + \gamma_3(b)) + 2\zeta(2)(\gamma_1(c) + \gamma_1(b)) + \zeta(3)(\gamma_0(c) + \gamma_0(b)) - \zeta(2)^2 - \zeta(4). \quad (\text{A.4})$$

Here $\zeta(2)^2 + \zeta(4) = 7\zeta(4)/2$. The reflection $J_{10}(c) = J_{01}(1 - c)$ fixes the orientation of the indices. These are rederived base identities already covered by the incoming closure results; the extension is their systematic use with unequal dilations and the explicit scale terms.

The first two nontrivial contact polynomials are

$$b_{0,k}(L) = L - H_k, \quad (\text{A.5})$$

$$b_{1,k}(L) = \frac{L^2}{2} - H_k L + \frac{H_k^2 - H_k^{(2)}}{2}, \quad (\text{A.6})$$

$$b_{2,k}(L) = \frac{L^3}{3} - H_k L^2 + (H_k^2 - H_k^{(2)})L - \frac{H_k^3 - 3H_k H_k^{(2)} + 2H_k^{(3)}}{3}. \quad (\text{A.7})$$

The empty harmonic sums at $k = 0$ vanish. Thus the same formulas include the pure scale term $L^{n+1}/(n+1)$ without a separate convention.

A.3 The fifth and sixth correction polynomials

To make the corrected historical conjectures explicit, write h_j for $H_n^{(j)}$, including $h_1 = H_n$. The complete polynomials in Theorem 8.1 are

$$E_5(n) = h_1^5 - 10h_1^3 h_2 + 20h_1^2 h_3 + 15h_1 h_2^2 - 30h_1 h_4 - 20h_2 h_3 + 24h_5, \quad (\text{A.8})$$

$$E_6(n) = h_1^6 - 15h_1^4 h_2 + 40h_1^3 h_3 + 45h_1^2 h_2^2 - 90h_1^2 h_4 - 120h_1 h_2 h_3 + 144h_1 h_5 - 15h_2^3 + 90h_2 h_4 + 40h_3^2 - 120h_6. \quad (\text{A.9})$$

Therefore

$$\sum_{n \geq 1} \frac{E_5(n)}{(n+1)(n+2)} = 120, \quad \sum_{n \geq 1} \frac{E_6(n)}{(n+1)(n+2)} = 720, \quad (\text{A.10})$$

$$\begin{aligned} \sum_{n \geq 1} \frac{H_n E_5(n)}{(n+1)(n+2)} &= 120 \left(1 + \sum_{j=2}^6 \zeta(j) \right), \\ \sum_{n \geq 1} \frac{H_n E_6(n)}{(n+1)(n+2)} &= 720 \left(1 + \sum_{j=2}^7 \zeta(j) \right). \end{aligned} \quad (\text{A.11})$$

In the last numerator, the term arising from $45h_1^2 h_2^2$ is exactly $45H_n^3 (H_n^{(2)})^2$. The final term of E_6 is exactly $-120H_n^{(6)}$. These are the two positions requiring harmonic-order corrections in the printed formulas.

For arbitrary r, t , the finite coefficient expression used by the exact generator can be written without Gamma functions as

$$\frac{\Gamma(1-u-v)}{\Gamma(2-u)\Gamma(2-v)} = \frac{1}{(1-u)(1-v)} \exp \left(\sum_{j=2}^{\infty} \frac{\zeta(j)}{j} ((u+v)^j - u^j - v^j) \right). \quad (\text{A.12})$$

Only terms through total degree $r+t$ are required for the coefficient $r!t![u^r v^t]$. When $r=t=0$, the complete $n \geq 0$ sum is 1 and the $n \geq 1$ sum is 1/2; for all positive total orders the $n=0$ term vanishes. Recording this exception prevents another normalization error at the first coefficient.

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